

# Invariant measures for the nonlinear stochastic heat equation with no drift term

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August 26, 2022

## Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation  $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ , where  $b$  is assumed to be a globally Lipschitz continuous function and the noise  $\dot{W}$  is a centered and spatially homogeneous Gaussian noise that is white in time. Using the martingale theory or the random field approach for stochastic partial differential equations laid out by John Walsh and Robert Dalang, we identify a set of conditions on the initial data, the correlation measure and the weight function  $\rho$ , which will together guarantee the existence of an invariant measure in the weighted space  $L^2_\rho(\mathbb{R}^d)$ . In particular, our result allows the *parabolic Anderson model* (i.e., the case when  $b(u) = \lambda u$ ) starting from the Dirac delta measure.

*MSC 2010 subject classification:* 60H15, 60H07, 60F05.

*Keywords:* Stochastic heat equation, parabolic Anderson model, invariant measure, Dirac delta initial condition, weighted  $L^2$  space, Matérn class of correlation functions, Bessel kernel.

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# 1 Introduction

In this paper, we study the following *nonlinear stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot). \end{cases} \quad (1.1)$$

The noise,  $\dot{W}(t, x)$ , is a centered Gaussian noise that is white in time and homogeneously colored in space defined on a complete probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  with the natural filtration  $\{\mathcal{F}_t\}_{t \geq 0}$  generated by the noise. Its covariance structure is given by

$$J(\psi, \phi) := \mathbb{E}[W(\psi)W(\phi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\psi(s, \cdot) * \tilde{\phi}(s, \cdot))(x), \quad (1.2)$$

where  $\psi$  and  $\phi$  are continuous and rapidly decreasing functions,  $\tilde{\phi}(x) := \phi(-x)$ , "\*" refers to the convolution in the spatial variable, and  $\Gamma$  is a nonnegative and nonnegative definite tempered measure on  $\mathbb{R}^d$  that is commonly referred to as the *correlation measure*. The Fourier transform of  $\Gamma$  (in the generalized sense) is also a nonnegative and nonnegative definite tempered measure, which is usually called the *spectral measure* and is denoted by  $\hat{f}(d\xi)$ <sup>1</sup>. Moreover, in the case where  $\Gamma$  has a density  $f$ , namely,  $\Gamma(dx) = f(x)dx$ , then  $\hat{f}(d\xi) = \hat{f}(\xi)d\xi$ .

The initial condition,  $\mu$ , is a deterministic, locally finite, regular, signed Borel measure that satisfies the following integrability condition at infinity:

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) |\mu|(dx) < \infty \quad \text{for all } a > 0, \quad (1.3)$$

where  $|\mu| = \mu_+ + \mu_-$  and  $\mu = \mu_+ - \mu_-$  refers to the *Hahn decomposition* of the measure  $\mu$ . Initial condition of this type, introduced in [7] and further explored in [9, 8], is called *rough initial condition*.

The function  $b(x, u)$  is uniformly bounded in the first variable and Lipschitz continuous in the second variable, i.e., for some constants  $L_b > 0$  and  $L_0 \geq 0$ ,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R} \text{ and } x \in \mathbb{R}^d. \quad (1.4)$$

In particular, our assumption allows the linear case  $b(x, u) = \lambda u$ , which is usually referred to as the *parabolic Anderson model* (PAM); see [4].

The SPDE (1.1) is understood in its *mild form*:

$$u(t, x) = J_0(t, x; \mu) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy), \quad (1.5)$$

where  $G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{1}{2t}|x|^2\right)$  is the heat kernel,

$$J_0(t, x) = J_0(t, x; \mu) := (G(t, \cdot) * \mu)(x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) \quad (1.6)$$

is the solution to the homogeneous equation, and the stochastic integral is the *Walsh integral*. We refer the interested readers to [25, 13, 12, 9] for more details of this setup.

The aim of this paper is to investigate the conditions required to guarantee the existence of an invariant measure for the solution to (1.1), which is a crucial step towards the study

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<sup>1</sup>See Remark 2.1 for the convention of Fourier transform that is used in this paper.

of the ergodicity of the system. The general procedure has been laid out by Tessitore and Zabczyk [24], which essentially involves two parts: firstly one needs to show that the laws of the solution to (1.1) form a family of *Markovian transition functions* on some Hilbert space,  $H$ , and the corresponding *Markovian semigroup* is Feller; and secondly one needs to establish that the moments of solution is bounded in time (see (1.12) below). For the second point, it requires some substantial work (see Theorems 1.2 and 1.4 below). On the other hand, the first point has been shown to be true for our case of interest (see, e.g., [11, Chapter 9]) with the following weighted  $L^2(\mathbb{R}^d)$  space as our underlying Hilbert space as in [24]:

**Definition 1.1.** A function  $\rho : \mathbb{R}^d \mapsto \mathbb{R}$  is called an *admissible weight function* if it is a strictly positive, bounded, continuous, and  $L^1(\mathbb{R}^d)$ -integrable function such that for all  $T > 0$ , there exists a constant  $C_\rho(T)$  such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d. \quad (1.7)$$

Moreover, we denote by  $L_\rho^2(\mathbb{R}^d)$  the corresponding Hilbert space of  $\rho$ -weighted square integrable functions and we use  $\langle \cdot, \cdot \rangle_\rho$  and  $\|\cdot\|_\rho$  to denote the inner product and norm on  $L_\rho^2(\mathbb{R}^d)$ :

$$\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x)dx \quad \text{and} \quad \|f\|_\rho := \left( \int_{\mathbb{R}^d} |f(x)|^2 \rho(x)dx \right)^{1/2}.$$

From now on we will denote  $H := L_\rho^2(\mathbb{R}^d)$ . In Corollary 1.3 below, we will see that the solution to (1.1) maps  $H$  into  $H$  in the sense that for each  $\zeta \in H$  and for all  $t > 0$ , we have the function  $x \mapsto u(t, x; \zeta)$  almost surely in  $H$ , where we use the notation  $u(t, x; \mu)$  to denote the solution to (1.1) starting from  $\mu$ . Moreover, we introduce a new condition, namely (1.20), which shows that the solution to (1.1) may also map measures that are outside of  $H$  into  $H$ . Thus, this condition allows us to consider more general initial conditions while at the same time still operate in the same framework laid out by the authors in [24, 11]. We provide some examples of permissible initial conditions that are outside of  $H$  in Section 2 below.

A probability measure,  $\eta$ , on the Borel  $\sigma$ -field of  $H$ , which we denote as  $\mathcal{B}(H)$ , is said to be *invariant* for (1.1) if

$$\eta(A) = \int_H \mathcal{L}(u(t, \cdot; \zeta))(A) \eta(d\zeta), \quad \text{for all } t \geq 0 \text{ and } A \in \mathcal{B}(H), \quad (1.8)$$

where  $\mathcal{L}(u(t, \cdot; \mu))(A)$  denotes the law of the solution:

$$\mathcal{L}(u(t, \cdot; \mu))(A) = \mathbb{P}[\omega \in \Omega : u(t, \cdot; \mu)(\omega) \in A], \quad A \in \mathcal{B}(H).$$

For notation, let us denote  $P_t(\zeta, A) := \mathcal{L}(u(t, \cdot; \zeta))(A)$  where  $\zeta \in H$  and  $A \in \mathcal{B}(H)$ . Then we can define the Markovian transition semi-group corresponding to  $P_t$  as

$$P_t \varphi(\zeta) = \int_H \varphi(y) P_t(\zeta, dy), \quad \text{for all } \varphi \in C_b(H),$$

where  $C_b(H)$  denotes the space of bounded continuous functions on  $H$ . Moreover, if we define  $\mathcal{M}(H)$  as the space of probability measures on  $H$ , then for each  $\nu \in \mathcal{M}(H)$ , we define the measure  $P_t^* \nu \in \mathcal{M}(H)$  as follows:

$$P_t^* \nu(A) = \int_H P_t(y, A) \nu(dy), \quad \text{for all } A \in \mathcal{B}(H),$$

and  $P_t^*$  can be shown to be the adjoint of  $P_t$  in the sense that for all  $\varphi \in C_b(H)$  and  $\nu \in \mathcal{M}(H)$ ,

$$\langle \varphi, P_t^* \nu \rangle = \langle P_t \varphi, \nu \rangle$$

where,

$$\langle \varphi, \nu \rangle := \int_H \varphi(y) \nu(dy) \quad \text{for } \varphi \in H \text{ and } \nu \in \mathcal{M}(H).$$

It is known that for each  $H$ -valued and  $\mathcal{F}_0$  measurable random variable,  $\zeta$ , with law  $\nu \in \mathcal{M}(H)$ , that  $P_t(\zeta, \cdot) = P_t^* \nu(\cdot)$  and under the spacial case that  $\zeta \in H$  is deterministic, then  $P_t(\zeta, \cdot) = P_t^* \delta_\zeta(\cdot)$  (see [11, Propostion 11.1]). Moreover, by [11, Proposition 11.2], if one can show that there exists a  $\zeta \in H$ , either random or deterministic, and a  $\eta \in \mathcal{M}(H)$  such that for each  $\varphi \in C_b(H)$ , the following limit exists:

$$\langle \varphi, P_t^* \nu \rangle \rightarrow \langle \varphi, \eta \rangle, \quad \text{as } t \rightarrow \infty, \quad (1.9)$$

where  $\nu \in \mathcal{M}(H)$  is the law of  $\zeta$ , then  $\eta$  is invariant for  $P_t$ . With this said, it would be interesting to show whether  $\eta$  in (1.9) is unique in the sense that for all  $\nu \in \mathcal{M}(H)$ , (1.9) holds true for  $\eta$  fixed. According to [11, Propostion 11.4], in order to show this uniqueness, it suffices to consider only deterministic  $\zeta \in H$ , in other words, it suffices to show that there exists an  $\eta$  such that for all  $\varphi \in C_b(H)$  and for all deterministic  $\zeta \in H$ , the following limit holds true:

$$\langle \varphi, P_t^* \delta_\zeta \rangle \rightarrow \langle \varphi, \eta \rangle, \quad \text{as } t \rightarrow \infty. \quad (1.10)$$

However, Proposition 2.9 below suggests that the limit in (1.10) may not exist for all  $\zeta \in H$ , which implies that (1.9) may not exist for some  $\nu \in \mathcal{M}(H)$ . Thus uniqueness of the invariant measure,  $\eta$ , in this situation would resort to proving that for all pairs  $(\nu, \tilde{\eta}) \in \mathcal{M}(H) \times \mathcal{M}(H)$  such that the limit in (1.9) exists,  $\tilde{\eta} = \eta$  in each pair. Moreover, even verifying (1.10) for one particular  $\zeta \in H$  is challenging and this has not yet been done for the stochastic heat equation as in (1.1). Instead, the common strategy, which we will follow in this paper, is to resort to the *Krylov-Bogoliubov theorem* (see, e.g., [11, Theorem 11.7]), and this theorem implies that once the tightness of  $\{\mathcal{L}(u(t, \cdot; \mu))\}_{t \geq t_0}$  for some  $t_0 \geq 0$  is established, one can construct an invariant measure via

$$\eta(A) = \lim_{n \rightarrow \infty} \frac{1}{T_n} \int_{t_0}^{T_n+t_0} \mathcal{L}(u(t, \cdot; \mu))(A) dt, \quad (1.11)$$

for some sequence  $\{T_n\}_{n \geq 1}$  with  $T_n \uparrow \infty$ . A critical step in obtaining the tightness is to show that the following moment is uniformly bounded in time:

$$\sup_{t \geq 0} \mathbb{E} \left( \|u(t, \cdot)\|_\rho^2 \right) < \infty, \quad (1.12)$$

and this will be the content of our Theorem 1.4 below.

In the literature, the existence of invariant measure of the stochastic heat equation is more commonly studied with a drift term and/or with the diffusion coefficient  $b(\cdot)$  being bounded; see Section 2.1 for a brief account. On the contrary, the existence of an invariant measure under the settings of equation (1.1) has been rarely studied. To the best of our knowledge, this current article and the one by Tessitore and Zabczyk [24] are the only papers that consider the case where the spatial domain is the whole space  $\mathbb{R}^d$ , the diffusion term,  $b(x, u)$ , is globally Lipschitz in the second variable uniformly in the first variable (see (1.4)), and there is no additional drift term. One of the challenges is to identify the minimal conditions so that the probability moments of the solution are bounded in time (see (1.12)). It is known that the solution to (1.1) is usually *intermittent*, namely, its moments possess a certain exponential growth in  $t$ ; see, e.g., [4, 16]. For that reason, one has to impose some additional assumptions either on the

initial conditions, or the noise, or the coefficients of (1.1), or all of them in order to combat this exponential growth in time of the moments.

As Tessitore and Zabczyk [24] is the only other paper that we know of that considers the same set up as equation (1.1), it has a strong influence in the current work. From a purely mathematical stand point, something interesting about this work is that all prior articles cited relating to the existence of invariant measures, including [24], use the theory of the stochastic integral formulated by Da Prato and Zabczyk [11], while in this paper we formulate and solve the problem using the random field language of Walsh [25] with some state-of-the-art moment estimates. Even though both theories have been shown to be more or less equivalent [14], they appear quite different from a technical sense. Thus, initiating a crossover from one theory to the other, as this paper does, is appealing on its own.

The improvements over [24] consist of the following aspects.

1. Results in [24] allow essentially all bounded functions as the initial conditions, though Theorem 3.3 (*ibid.*) was proved only for the constant one initial condition. Here we give precise conditions on the initial condition, namely, (1.23), which allows a much wider class of initial conditions, including unbounded functions and measures such as the Dirac delta measure; see Examples 2.7 and 2.8. We emphasize that the Dirac delta initial measure plays a very prominent role in the study of the stochastic heat equation; see, e.g., [1].
2. We give a more easily verifiable condition in (1.13a) on the spectral density  $\hat{f}$  and provide examples of suitable  $\hat{f}$  in Section 2.4.

We now will introduce some conditions and quantities that will be needed for our main results. In order to have moments bounded in time, one has to first identify the exact conditions under which the moments to (1.1) do not possess exponential growth. This question has been answered in [9, Theorem 1.3 and Lemma 2.5], where necessary and sufficient conditions are given. More precisely, for bounded second moments, one has to have the spatial dimension  $d \geq 3$ , and in addition, the spectral measure  $\hat{f}$  and Lipschitz constant  $L_b$  of  $b(\cdot)$  have to satisfy the following two conditions:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad (1.13a)$$

$$\text{and } 64L_b^2 < \frac{1}{2\Upsilon(0)}. \quad (1.13b)$$

Moreover, these two conditions will guarantee the existence of the following non-empty open interval:

$$(2^7 L_b^2 \Upsilon(0), 1) \neq \emptyset. \quad (1.14)$$

Note that condition (1.13a) is a strengthened version of *Dalang's condition*:

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty, \quad \text{for some (and hence) all } \beta > 0. \quad (1.15)$$

Recall that in order to obtain the Hölder continuity of the solution, one needs to strengthen (1.15) in a different way. Indeed, what is required is that for some  $\alpha \in (0, 1]$ ,

$$\Upsilon_\alpha(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{(\beta + |\xi|^2)^{1-\alpha}} < \infty \quad \text{for some (hence all) } \beta > 0; \quad (1.16)$$

see Theorem 1.8 of [8] or [22]. Similarly, one can further strengthen condition (1.16) to

$$\Upsilon_\alpha(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^{2(1-\alpha)}} < \infty \quad \text{for some } \alpha \in (0, 1]. \quad (1.17)$$

We use the convention that when  $\alpha = 0$ , we simply drop it from the expression  $\Upsilon_\alpha(\beta)$ , i.e.,  $\Upsilon(\beta) = \Upsilon_0(\beta)$ . The relations of these conditions are illustrated in Figure 1.

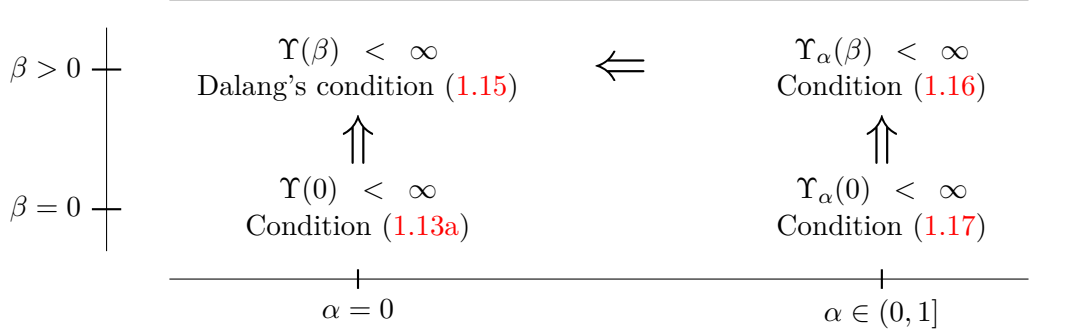


Figure 1: Relations among conditions (1.15), (1.16), (1.17) and (1.13a). Check also Lemma 4.5 for the relation between  $\Upsilon(0) < \infty$  and  $\Upsilon_\alpha(0) < \infty$ .

We will also need the following slightly different condition:

$$\mathcal{H}_{\alpha/2}(t) < \infty \quad \text{for some } \alpha \in (0, 1] \text{ and for all } t > 0, \quad (1.18)$$

where

$$\mathcal{H}_\alpha(t) := \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2). \quad (1.19)$$

The quantity  $\mathcal{H}_\alpha(t)$  will appear naturally in the proof of Lemma 4.2, however, as is shown in Lemma 4.5 below, condition (1.17) will imply condition (1.18). Moreover, when we assume (1.13a) holds true, which will always be the case, then these two conditions become equivalent. The quantity  $\Upsilon_\alpha(0)$  has the benefit of being more compactly defined as well as being more well recognized.

We are now ready to state our two main results of the paper: Theorems 1.2 and 1.4. These theorems will be proved in Sections 3 and 5 respectively.

**Theorem 1.2.** *Let  $u(t, x; \mu)$  be the solution to (1.1) starting from  $\mu$  which satisfies (1.3). Assume that*

- (i)  $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$  is a nonnegative  $L^1(\mathbb{R}^d)$  function;
- (ii) for all  $t > 0$ , the initial condition  $\mu$  satisfies  $\mathcal{G}_\rho(t; |\mu|) < \infty$  where

$$\mathcal{G}_\rho(t; \mu) := \int_{\mathbb{R}^d} J_0^2(t, x; \mu) \rho(x) dx; \quad (1.20)$$

- (iii) the spectral measure  $\hat{f}$  and the Lipschitz constant  $L_b$  satisfy the two conditions in (1.13).

Then there exists an unique  $L^2(\Omega)$ -continuous solution  $u(t, x)$  such that for some constant  $C > 0$ , which does not depend on  $t$ , the following holds:

$$\mathbb{E} \left( \|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; \mu^*) < \infty, \quad \text{for any } t > 0, \quad (1.21)$$

where  $\mu^* := 1 + |\mu|$ .

We now state and prove a quick Corollary which shows that the solution to (1.1) starting from a  $L^2_\rho(\mathbb{R}^d)$  initial condition will almost surely be in  $L^2_\rho(\mathbb{R}^d)$  for all  $t > 0$ .

**Corollary 1.3.** *Under the same assumptions of Theorem 1.2, if in addition  $\rho$  is admissible (see Definition 1.1), then the solution  $u(t, \cdot; \zeta)$  to (1.1) almost surely exists in  $L^2_\rho(\mathbb{R}^d)$  for all  $t > 0$ , whenever the initial condition is also in  $L^2_\rho(\mathbb{R}^d)$ , i.e.,  $\zeta \in L^2_\rho(\mathbb{R}^d)$ .*

*Proof.* Choose and fix an arbitrary  $\zeta \in L^2_\rho(\mathbb{R}^d)$  and set  $\zeta^* = 1 + |\zeta|$ . It is clear that  $\zeta^* \in L^2_\rho(\mathbb{R}^d)$ . By (ii) of Theorem 1.2, it suffices to show the finiteness of  $\mathcal{G}_\rho(t; \zeta^*)$  for all  $t > 0$ . Indeed, by Hölder's inequality,

$$\mathcal{G}_\rho(t; \zeta^*) = \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} G(t, x - y) \zeta^*(y) dy \right)^2 \rho(x) dx \leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G(t, x - y) \zeta^*(y)^2 \rho(x) dy dx.$$

Now for any  $t > 0$ , choose  $T > t$  and let  $C_\rho(T)$  be as in (1.7). Then,

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G(t, x - y) \zeta^*(y)^2 \rho(x) dy dx = \int_{\mathbb{R}^d} (G(t, \cdot) * \rho)(y) |\zeta(y)|^2 dy \leq C_\rho(T) \|\zeta^*\|_\rho^2 < \infty.$$

Since  $t$  and  $T$  are arbitrary, this completes the proof of the corollary.  $\square$

**Theorem 1.4.** *Let  $u(t, x)$  be the solution to (1.1) starting from  $\mu$  and let  $\rho$  be an admissible weight function. Assume that*

(i) *there exists another admissible weight  $\tilde{\rho}$  such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\tilde{\rho}(x)} dx < \infty; \quad (1.22)$$

(ii) *the weight function  $\tilde{\rho}$  and the initial condition satisfy the following condition:*

$$\limsup_{t>0} \mathcal{G}_{\tilde{\rho}}(t; |\mu|) < \infty; \quad (1.23)$$

(iii) *the spectral measure  $\hat{f}$  and the Lipschitz constant  $L_b$  satisfy the two conditions in (1.13);*

(iv) *for some  $\alpha \in (2^7 \Upsilon(0) L_b^2, 1)$  (see (1.14)), the spectral measure  $\hat{f}$  satisfies (1.17).*

*Then we have that*

(a) *for any  $\tau > 0$ , the sequence of laws of  $\{\mathcal{L}u(t, \cdot; \mu)\}_{t \geq \tau}$  is tight, i.e., for any  $\epsilon \in (0, 1)$ , there exists a compact set  $\mathcal{K} \subset L^2_\rho(\mathbb{R}^d)$  such that*

$$\mathcal{L}u(t, \cdot; \mu)(\mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau > 0; \quad (1.24)$$

(b) *there exists an invariant measure for the laws  $\{\mathcal{L}u(t, \cdot; \mu)\}_{t>0}$  in  $L^2_\rho(\mathbb{R}^d)$ .*

This paper is organized as follows: In Section 2, we further discuss our main results and provide some examples. In particular, in Section 2.5 we will extend the discussion about the weight functions; in Section 2.3 we show that our results could include a wider class of initial conditions; in Section 2.4 we give some explicit examples of the correlation functions that satisfy (1.17); in Section 2.2 we compare our conditions with those obtained by Tessitore and Zabczyk [24]. The two main Theorems 1.2 and 1.4 will be proved in Sections 3 and 5, respectively.

We conclude this introduction by introducing some notation and formulas that we use throughout the paper. We will use  $\|X\|_p$  to denote the  $L^p(\Omega)$  norm, namely,  $\|X\|_p = (\mathbb{E}(|X|^p))^{1/p}$ . We will also use the following factorization property of the heat kernel,

$$G(t, x)G(s, y) = G\left(\frac{ts}{t+s}, \frac{sx+ty}{t+s}\right) G(t+s, x-y), \quad (1.25)$$

which can be easily verified and has been used extensively and critically in [7, 9, 8]. Next, we will need the following spherical coordinate integration formula:

$$\int_{\mathbb{R}^d} f(|x|)dx = \sigma(\mathbb{S}^{d-1}) \int_0^\infty f(r)r^{d-1}dr,$$

where  $\sigma(\mathbb{S}^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$  and  $\Gamma(x)$  denotes the Gamma function. We use " $\sim$ " to denote the standard asymptotic equivalent relation. Lastly, the convention of Fourier transform is given in Remark 2.1 below.

## 2 Discussion and examples

### 2.1 SHE with a drift term

When present, authors usually denote the drift term as  $f(x, u(t, x))$ , however, we have already used  $f$  to denote the correlation function in this paper, so we will use  $g$  for the drift.

In the literature, the existence of invariant measure of the stochastic heat equation is more commonly studied with a drift term  $g(x, u(t, x))$ , namely,

$$\left(\frac{\partial}{\partial t} - \frac{1}{2}\Delta\right)u(t, x) = g(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) \quad x \in \mathcal{O}, t > 0, \quad (2.1)$$

For example, to cancel the growth of the moments, the authors in [2, 5, 6] require a drift coefficient,  $g(u) : \mathbb{R} \rightarrow \mathbb{R}$ , to be added to the right hand side of equation (1.1) and that this term must satisfy the following dissipativity condition for some  $m, k_i, c_i > 0$  as  $|u| \rightarrow \infty$ :

$$\begin{cases} g(u) \leq -k_1|u|^m + k_2 & u > 0 \\ g(u) \geq c_1|u|^m - c_2 & u < 0. \end{cases} \quad (2.2)$$

Furthermore, Cerrai [5, 6] considered the case of a bounded spatial domain while Assing and Manthey [2] considered the whole space  $\mathbb{R}^d$ . In addition, the just mentioned papers only allow for bounded initial conditions, however, an unbounded Lipschitz continuous diffusion term as in (1.4) is allowed to be considered. It is also worth noting that Eckmann and Hairer [15] also require the above dissipativity condition (2.2) along with a bounded initial condition, but they consider the case of an additive noise.

Several works which do not require an added drift term with dissipativity as in (2.2) include Misiats *et al* [20, 19]. In the [20, Theorem 1.2], they provide a result guaranteeing the existence of an invariant measure for the stochastic heat equation on the whole space  $\mathbb{R}^d$ . More precisely, they allow for a drift term,  $g(x, u) : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ , such that for all  $x \in \mathbb{R}^d$  and  $u_1, u_2 \in \mathbb{R}$ ,

$$|g(x, 0)| \leq \varphi(x) \quad \text{and} \quad |g(x, u_1) - g(x, u_2)| \leq L\varphi(x)|u_1 - u_2|,$$

for some  $L > 0$  where  $\varphi(x)$  must decay fast enough such that  $\varphi/\sqrt{\rho} \in L^2(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d)$  and  $\rho$  is the admissible weight. Thus  $f = 0$  is allowed. Moreover, they require the diffusion term,  $b$ , to satisfy the following:

$$|b(x, u_1) - b(x, u_2)| \leq L\varphi(x)|u_1 - u_2|,$$

which excludes the parabolic Anderson model. Lastly, Theorem 1.2 *ibid.* requires the initial condition to be in  $L^2(\mathbb{R}^d)$  and this excludes the popular cases of  $u(0, x) = 1$  and  $u(0, \cdot) = \delta_0(\cdot)$ , where  $\delta_0$  is the Dirac delta distribution. Our Theorem 1.4 includes both of these initial conditions and we show this in Section 2.

## 2.2 The conditions for the spectral measures by Tessitore and Zabczyk

Under the setting of the whole space,  $\mathbb{R}^d$ , Tessitore and Zabczyk [24] prove the existence of an invariant measure for (1.1) in  $L^2_\rho(\mathbb{R}^d)$  under the assumption that there exists a  $\varphi \in L^2_\rho(\mathbb{R}^d) \cap L^2_\rho(\mathbb{R}^d)$  where  $\rho(\tilde{\rho})^{-1} \in L^1(\mathbb{R}^d)$  and the solution starting from  $\varphi$  is bounded in probability in  $L^2_\rho(\mathbb{R}^d)$  and also that the spectral density  $\hat{f}$  satisfies

$$\hat{f} \in L^p(\mathbb{R}^d) \quad \text{where} \quad \frac{d-2}{d} < \frac{1}{p}; \quad (2.3)$$

see Hypothesis 2.1 (*ibid.*). However as was illustrated in Theorem 3.3 (*ibid.*), in order to apply this theorem to a specific initial condition in  $L^2_\rho$  (or to have moments uniformly bounded in time), the following additional assumptions were imposed:

$$d \geq 3 \quad \text{and} \quad L_b^{-2} > \frac{\Gamma(d/2 - 1)2^{d/2-2}}{(2\pi)^{2d}} \int_{\mathbb{R}^d} \left( \left| \mathcal{F}(\sqrt{\hat{f}}) \right| * \left| \mathcal{F}(\sqrt{\hat{f}}) \right| \right) (\xi) |\xi|^{2-d} d\xi, \quad (2.4)$$

where the convention of the Fourier transform is given in Remark 2.1. With these assumptions, they were able to prove that (1.1) starting from the constant 1 satisfies (1.12) and thus is bounded in probability, verifying the existence of an invariant measure. Moreover, the measure takes the form of (1.11) above with  $\mu = 1$ . Lastly we mention that due to its difficult nature, (2.4) was not calculated for a specific  $\hat{f}$  (see Examples 2.3 and 2.5 below).

**Remark 2.1.** The Fourier transform may be defined differently depending on how one would like to handle the  $2\pi$  constant. In this paper (as in [9, 8]), we use the convention that

$$\hat{\phi}(\xi) = \mathcal{F}\phi(\xi) := \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi. \quad (2.5)$$

Hence, Plancherel's theorem takes the form of

$$\int_{\mathbb{R}^d} \psi(x) \phi(x) dx = (2\pi)^{-d} \int_{\mathbb{R}^d} \hat{\psi}(\xi) \overline{\hat{\phi}(\xi)} d\xi. \quad (2.6)$$

Note that the authors in [24] did not explicitly mention their convention of the Fourier transform. However, the proof of Theorem 3.3 (*ibid.*) suggests that the following convention has been used:

$$\hat{\phi}(\xi) = \mathcal{F}\phi(\xi) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi.$$

Hence, Plancherel's theorem takes the form,  $\int_{\mathbb{R}^d} \psi(x) \phi(x) dx = \int_{\mathbb{R}^d} \hat{\psi}(\xi) \overline{\hat{\phi}(\xi)} d\xi$ , without the additional factor  $(2\pi)^{-d}$ . In particular, the spectral density  $\gamma$  (*ibid.*) corresponds to  $(2\pi)^{-d/2} \hat{f}$  in this paper. Our equation (2.4), which is condition (3.4) (*ibid.*), takes into account this difference, therefore explaining the slightly different factor in front of the integral in (2.4) from that in (3.4) (*ibid.*).

Recall the definition of  $\mathcal{F}^{-1}$  in (2.5) above. We may rewrite the integral in the second condition of (2.4) as follows:

$$\Gamma(d/2 - 1)2^{d/2-2} \int_{\mathbb{R}^d} \left( \left| \mathcal{F}^{-1}(\sqrt{\widehat{f}}) \right| * \left| \mathcal{F}^{-1}(\sqrt{\widehat{f}}) \right| \right)(x) |x|^{2-d} dx.$$

We now consider a couple of cases and give two examples that illustrate some of the difficulties that may arise because of the square root and absolute value. Before that, let us first introduce a nonnegative and nonnegative definite function in  $\mathbb{R}$ :

**Example 2.2.** Let  $d = 1$  and  $g(x) = \frac{1}{2}1_{[-1,1]}(x)$ . Then it is clear that  $\widehat{g}(\xi) = \xi^{-1} \sin(\xi)$ . Now set  $f(x) = (g * g)(x) = 2^{-2} \max(2 - |x|, 0)$ . It is clear that  $f$  is nonnegative. It is also nonnegative-definite because  $\widehat{f}(\xi) = \widehat{g}(\xi)^2 = \xi^{-2} \sin^2(\xi) \geq 0$ .

### 2.2.1 Case I

In this case, we assume that

$$\mathcal{F}^{-1}(\sqrt{\widehat{f}}) \geq 0 \text{ and } \widehat{f} > 0 \text{ on } \mathbb{R}^d. \quad (2.7)$$

This is not a trivial assumption (see Examples 2.3 and 2.4 below), however one can check that the Bessel kernels satisfy this condition. Under (2.7), there is no ambiguity when taking the square root and one can remove the absolute value to see that

$$\begin{aligned} \Gamma(d/2 - 1)2^{d/2-2} \int_{\mathbb{R}^d} \left( \mathcal{F}^{-1}(\sqrt{\widehat{f}}) * \mathcal{F}^{-1}(\sqrt{\widehat{f}}) \right)(x) |x|^{2-d} dx \\ = \frac{1}{(2\pi)^{3d/2}} \int_{\mathbb{R}^d} \widehat{f}(\xi) |\xi|^{-2} d\xi = (2\pi)^{d/2} \Upsilon(0), \end{aligned}$$

where we have applied Plancherel's theorem (see (2.6)) and the Fourier transform for Riesz kernel (in the generalized sense):

$$\mathcal{F}(|\cdot|^{-\alpha})(\xi) = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} |\xi|^{-(d-\alpha)}, \quad \text{for } \alpha \in (0, d) \text{ and } \xi \in \mathbb{R}^d.$$

Hence, the second condition in (2.4) can be equivalently written as  $L_b^{-2} > (2\pi)^{d/2} \Upsilon(0)$ . Comparing this with (1.13b), namely,  $L_b^{-2} > 128 \Upsilon(0)$ , our condition is sharper when  $d > \frac{14 \log(2)}{\log(2\pi)} \approx 5.28$ .

The following example shows that  $\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$  may not be non-negative even if  $f$  is non-negative and non-negative definite, thus simply removing the absolute values in (2.4) is no longer possible.

**Example 2.3.** Suppose that  $d = 1$  and  $\widehat{f}(\xi) = 2^{-2} \max\{2 - |\xi|, 0\}$ . Then as explained in Example 2.2, both  $f$  and  $\widehat{f}$  are non-negative and non-negative definite and  $f(x) = (2\pi)^{-1} x^{-2} \sin^2(x)$ . We claim that

$$\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x) \text{ takes both positive and negative values.} \quad (2.8)$$

Indeed, for all  $x \in \mathbb{R}$ ,

$$\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x) = \frac{2}{2\pi} \int_0^2 2^{-1} \sqrt{2-\xi} \cos(x\xi) d\xi = \frac{1}{\pi} \int_0^{\sqrt{2}} \xi^2 \cos(x(2-\xi^2)) d\xi$$

$$\begin{aligned}
&= \frac{1}{\pi} \cos(2x) \int_0^{\sqrt{2}} \xi^2 \cos(x\xi^2) d\xi + \frac{1}{\pi} \sin(2x) \int_0^{\sqrt{2}} \xi^2 \sin(x\xi^2) d\xi \\
&= -\frac{1}{\pi} \cos(2x) \int_0^{\sqrt{2}} \frac{\sin(x\xi^2)}{x} d\xi + \frac{1}{\pi} \sin(2x) \int_0^{\sqrt{2}} \frac{\cos(x\xi^2)}{x} d\xi,
\end{aligned}$$

where we have applied the change of variables  $\xi' = \sqrt{2-\xi}$  in the second equality and in the last equality we have applied an integration by parts. By using the *Fresnel integrals* (see, e.g., [21, 7.2 (iii)])

$$\mathcal{S}(z) = \int_0^z \sin\left(\frac{\pi t^2}{2}\right) dt \quad \text{and} \quad \mathcal{C}(z) = \int_0^z \cos\left(\frac{\pi t^2}{2}\right) dt,$$

the above expression can be further simplified via the change of variables  $\xi' = \sqrt{2|x|/\pi} \xi$  to

$$\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x) = (8\pi)^{-1/2} |x|^{-3/2} \left( -\cos(2|x|) \mathcal{S}\left(\frac{2\sqrt{|x|}}{\sqrt{\pi}}\right) + \sin(2|x|) \mathcal{C}\left(\frac{2\sqrt{|x|}}{\sqrt{\pi}}\right) \right), \quad x \in \mathbb{R}.$$

This proves the claim in (2.8); see also Figure 2 for some plots.

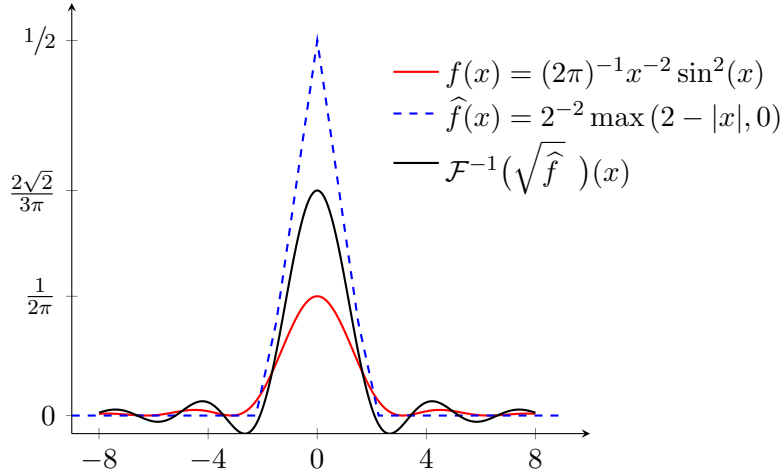


Figure 2: One example that  $\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$  assumes both positive and negative values.

The next example shows that the spectral density given in Example 2.3 can be easily handled by the conditions we presented in Theorem 1.4, namely showing the finiteness of  $\Upsilon_\alpha(0)$ .

**Example 2.4.** From the one-dimension example in Example 2.3, one can construct a  $d$ -dimensional counterpart. Here is one way to do that. Let  $f_1$  and  $\widehat{f}_1$  be the  $f$  and  $\widehat{f}$ , respectively, in Example 2.3. Then define

$$f_d(x) := \prod_{i=1}^d f_1(x_i), \quad x \in \mathbb{R}^d, \quad \text{and hence} \quad \widehat{f}_d(\xi) := \prod_{i=1}^d \widehat{f}_1(\xi_i), \quad \xi \in \mathbb{R}^d.$$

From Example 2.3, we see that  $\widehat{f}_d(\cdot)$  does not satisfy condition (2.7). In contrast, it is straightforward to verify that

$$\Upsilon_\alpha(0) = C \int_{\mathbb{R}^d} \frac{\prod_{i=1}^d \max\{2 - |\xi_i|, 0\}}{|\xi|^{2(1-\alpha)}} d\xi \leq C \int_{|\xi| \leq 2\sqrt{d}} \frac{2^d}{|\xi|^{2(1-\alpha)}} d\xi = C \int_0^{2\sqrt{d}} \frac{r^{d-1}}{r^{2(1-\alpha)}} dr.$$

Hence, if  $\alpha > 1 - d/2$ , then  $\Upsilon_\alpha(0) < \infty$ .

### 2.2.2 Case II

If  $\widehat{f}$  is not strictly positive, but only nonnegative, namely,  $\widehat{f}(\xi) = 0$  for some  $\xi \in \mathbb{R}^d$ , then removing the absolute value in (2.4) becomes tricky, which will be illustrated in the following example.

**Example 2.5.** Let  $f$  and  $g$  be given as Example 2.2. In this case,  $\widehat{f}(\cdot)$  is only nonnegative (not strictly positive) with infinitely many zeros. Hence, when taking the square root of  $\widehat{f}(\xi)$  as in (2.4), one needs to wisely choose the correct positive and negative branches:

1. Clearly, the signed version  $\sqrt{\widehat{f}(\xi)} = \xi^{-1} \sin(\xi)$  is preferable since its inverse Fourier transform can be easily computed, which is equal to  $g(x)$ . Moreover, because this inverse Fourier transform  $g(x)$  is clearly nonnegative, the absolute value signs in (2.4) do not pose any additional restrictions.
2. However, if one chooses the positive branches, namely,  $\sqrt{\widehat{f}(\xi)} = |\xi^{-1} \sin(\xi)|$ , then it is not clear how to compute its Fourier transform. In general, some bad choices of the positive/negative branches may make the conditions in (2.4) fail. For example, such choice may turn  $\sqrt{\widehat{f}(\xi)}$  into a distribution, and then taking the absolute value of a distribution (unless it is a measure) may be problematic. Another issue that may arise is when  $\sqrt{\widehat{f}(\xi)}$  is a well-defined function, taking on both positive and negative values and after taking the absolute value, the integral in (2.4) may blow up.

### 2.3 Various initial conditions

**Example 2.6** ( $L^\infty(\mathbb{R}^d)$  initial condition). We emphasize that if the initial condition,  $\mu$ , is deterministic and is such that  $\mu(dx) = \varphi(x)dx$  with  $\varphi \in L^\infty(\mathbb{R}^d)$ , then all conditions related to  $\mathcal{G}_\rho(\cdot)$  in both Theorems 1.2 and 1.4 are trivially satisfied. To be more precise, both Conditions (1.20) and (1.23) hold because

$$\mathcal{G}_\rho(t; |\varphi|) \leq \|\varphi\|_{L^\infty(\mathbb{R}^d)}^2 \|\rho\|_{L^1(\mathbb{R}^d)} < \infty \quad \text{uniformly for all } t \geq 0.$$

**Example 2.7** (Delta initial condition). In this example, we study the case when the initial condition,  $\mu$ , is the Dirac delta measure at zero, namely,  $\delta_0$ . Let  $\rho$  be a nonnegative  $L^1(\mathbb{R}^d)$  function. Since

$$\mathcal{G}_\rho(t; |\delta_0|) = \int_{\mathbb{R}^d} G(t, x)^2 \rho(x) dx \leq G(t, 0)^2 \|\rho\|_{L^1(\mathbb{R}^d)}, \quad \text{for all } t > 0,$$

we see that both conditions (1.20) and (1.23) are satisfied. In particular,  $\limsup_{t \geq 0} \mathcal{G}_\rho(t; \delta_0) = 0$ .

**Example 2.8** (More initial conditions not in  $L_\rho^2(\mathbb{R}^d)$ ). In this example, we study the case when  $\mu(dx) = |x|^{-\alpha} dx$  for some  $\alpha \in (0, d)$ . It is clear that when  $\alpha \in (d/2, d)$ ,  $\mu \notin L_\rho^2(\mathbb{R}^d)$ . However, in this case, we have

$$J_0(t, x) = (G(t, \cdot) * |\cdot|^{-\alpha})(x) \leq (G(t, \cdot) * |\cdot|^{-\alpha})(0).$$

On the other hand, Hence,

$$(G(t, \cdot) * |\cdot|^{-\alpha})(0) = \frac{2\pi^{d/2}}{\Gamma(d/2)} \times (2\pi t)^{-d/2} \int_0^\infty e^{-\frac{r^2}{2t}} r^{-\alpha+d-1} dr = C_* t^{-\alpha/2},$$

with  $C_* = 2^{-\alpha/2} \Gamma((d-\alpha)/2) / \Gamma(d/2)$ , which implies that

$$\mathcal{G}_\rho(t; |\cdot|^{-\alpha}) \leq \int_{\mathbb{R}^d} J_0^2(t, 0) \rho(x) dx = C_*^2 t^{-\alpha} \|\rho\|_{L^1(\mathbb{R}^d)}, \quad \text{for all } t > 0.$$

Therefore, we see that both conditions (1.20) and (1.23) are satisfied.

The following proposition shows that for initial conditions with unbounded tails, condition (1.20) may hold while condition (1.23) may fail.

**Proposition 2.9.** *Suppose that  $\rho(x) = \exp(-|x|)$ , which is an admissible weight function. Let the initial condition  $\mu$  be given as*

$$\mu(dx) = |x|^\alpha dx \quad \text{with } \alpha > 0.$$

Then for some constants  $C, C' > 0$  that depend on  $d$  and  $\alpha$ , it holds that

$$C'(1+t^\alpha) \leq \mathcal{G}_\rho(t; |\mu|) \leq C(1+t^\alpha), \quad \text{for all } t > 0. \quad (2.9)$$

In particular, this implies that condition (1.20) is satisfied, but condition (1.23) fails.

*Proof.* Notice that by scaling arguments,

$$\begin{aligned} \mathcal{G}_\rho(t; |\mu|) &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} G(t, x-y) |y|^\alpha dy \right)^2 e^{-|x|} dx \\ &= \int_{\mathbb{R}^d} t^\alpha \left( \int_{\mathbb{R}^d} G\left(1, \frac{x}{\sqrt{t}} - z\right) |z|^\alpha dz \right)^2 e^{-|x|} dx \\ &= \int_{\mathbb{R}^d} t^{\alpha+d/2} \left( \int_{\mathbb{R}^d} G(1, \xi - z) |z|^\alpha dz \right)^2 e^{-\sqrt{t}|\xi|} d\xi. \end{aligned}$$

In the following, let  $C_d, C_\alpha, C'_\alpha, C_{\alpha,d}$  and  $C'_{\alpha,d}$  be generic constants that may depend on  $\alpha$  and  $d$  and may change their value at each appearance.

*Upper bound:* Because

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \leq C_\alpha \int_{\mathbb{R}^d} G(1, z) (|\xi|^\alpha + |z|^\alpha) dz \leq C'_\alpha (1 + |\xi|^\alpha),$$

we see that

$$\begin{aligned} \mathcal{G}_\rho(t; \mu) &\leq C_\alpha \int_{\mathbb{R}^d} t^{\alpha+d/2} (1 + |\xi|^{2\alpha}) e^{-\sqrt{t}|\xi|} d\xi \\ &= C_{\alpha,d} \int_0^\infty t^{\alpha+d/2} (1 + r^{2\alpha}) e^{-\sqrt{t}r} r^{d-1} dr \\ &= C_{\alpha,d} (t^\alpha \Gamma(d) + \Gamma(d+2\alpha)) \\ &= C'_{\alpha,d} (1 + t^\alpha) < \infty. \end{aligned}$$

Hence, this proves the upper bound in (2.9).

*Lower bound:* Now we prove the lower bound in (2.9). Indeed,

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \geq \int_{\mathbb{R}^d} G(1, z) ||\xi| - |z||^\alpha dz$$

$$\begin{aligned}
&\geq C_d \int_0^\infty ||\xi| - x|^\alpha e^{-\frac{x^2}{2}} x^{d-1} dx \\
&\geq C_d \int_1^2 ||\xi| - x|^\alpha dx \\
&= \frac{C_d}{1+\alpha} \psi(|\xi|),
\end{aligned}$$

where, by considering three cases, we have

$$\begin{aligned}
\psi(r) &= \begin{cases} (2-r)^{\alpha+1} - (1-r)^{\alpha+1} & \text{if } 0 < r < 1, \\ (2-r)^{\alpha+1} + (r-1)^{\alpha+1} & \text{if } 1 \leq r \leq 2, \\ (r-1)^{\alpha+1} - (r-2)^{\alpha+1} & \text{if } r > 2, \end{cases} \\
&= \operatorname{sgn}(2-r)|r-2|^{\alpha+1} + \operatorname{sgn}(r-1)|r-1|^{\alpha+1}.
\end{aligned}$$

We claim that

$$\inf_{r \geq 0} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} > 0. \quad (2.10)$$

With (2.10), we have that

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \geq C_{\alpha, d} \sqrt{1 + |\xi|^{2\alpha}}.$$

Then, by the same arguments as above for the upper bound, we obtain the lower bound in (2.9). It remains to prove (2.10), which will be proved in three cases.

When  $r > 2$ , we see that

$$\begin{aligned}
\frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} &\geq C_\alpha \frac{(r-1)^{\alpha+1} - (r-2)^{\alpha+1}}{(1+r)^\alpha} \\
&\geq C_\alpha \frac{(r-1)^\alpha(r-1) - (r-1)^\alpha(r-2)}{(1+r)^\alpha} \\
&= C_\alpha \left( \frac{r-1}{1+r} \right)^\alpha = C_\alpha \left( 1 - \frac{2}{1+r} \right)^\alpha \geq C_\alpha \left( 1 - \frac{2}{3} \right)^\alpha.
\end{aligned}$$

Note that in the first inequality above, we have considered two cases:  $2\alpha \geq 1$  and  $2\alpha < 1$ . When  $2\alpha < 1$ , we have used the concavity of  $x^{2\alpha}$ , namely,  $(1+r^{2\alpha})^{2\alpha}/2 \leq ((1+r)/2)^{2\alpha}$ ; when  $2\alpha \geq 1$ , we have used the super-additiity of  $x^{2\alpha}$ : namely, that for  $a, b > 0$  that  $(a+b)^{2\alpha} \geq a^{2\alpha} + b^{2\alpha}$ . This shows that  $\inf_{r > 2} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} > 0$ .

When  $r \in (1, 2]$ , elementary calculations show that the minimum of  $\psi(r)$  is achieved at  $r = 3/2$ . Hence,

$$\inf_{r \in (1, 2]} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} \geq \frac{\psi(3/2)}{\sqrt{1+4^\alpha}} > 0.$$

Similarly, when  $r \in (0, 1]$ , by differentiation, one finds that the function  $\psi(r)$  is nonincreasing. Hence, the minimum is achieved at  $r = 1$ :

$$\inf_{r \in (0, 1]} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} \geq \frac{\psi(1)}{\sqrt{2}} > 0.$$

Combing the above three cases proves (2.10). This completes the proof of Proposition 2.9.  $\square$

## 2.4 Bessel and other related kernels

**Example 2.10** (Bessel kernel). Let  $f_s$  denote the Bessel kernel with a strictly positive parameter  $s > 0$ . It is known that (see, e.g., Section 1.2.2 of [17])

1.  $f_s(x) > 0$  for all  $x \in \mathbb{R}^d$  and  $\|f_s\|_{L^1(\mathbb{R}^d)} = 1$ ;
2. there exists a constant  $C(s, d) > 0$  such that

$$f_s(x) \leq C(s, d) \exp(-|x|/2) \quad \text{for } |x| \geq 2;$$

3. there exists a constant  $c(s, d) > 0$  such that

$$\frac{1}{c(s, d)} \leq \frac{f_s(x)}{H_s(x)} \leq c(s, d) \quad \text{for } |x| \leq 2, \quad \text{with}$$

$$H_s(x) = \begin{cases} |x|^{s-d} + 1 + O(|x|^{s-d+2}) & \text{for } 0 < s < d, \\ \log\left(\frac{2}{|x|}\right) + 1 + O(|x|^2) & \text{for } s = d, \\ 1 + O(|x|^{s-d}) & \text{for } s > d; \end{cases}$$

4. the Fourier transform of  $f_s$  is strictly positive:

$$\mathcal{F}f_s(\xi) = \frac{1}{(1 + |\xi|^2)^{s/2}}. \quad (2.11)$$

Note that one can use (2.11) as the definition of the Bessel kernel. Properties 1 and 4 ensure that  $f_s$  is a nonnegative and nonnegative-definite tempered measure for all  $s > 0$ .

**Example 2.11** (Matérn class of correlation functions). The *Matérn class of correlation functions* has been widely used in spatial statistics; one may check the recent work [18] for references. Following Section 2.10 of [23], this class of correlation functions is given by

$$K(x) = \phi \cdot (\alpha|x|)^\nu \mathcal{K}_\nu(\alpha|x|), \quad \text{for } x \in \mathbb{R}^d \text{ with } \phi > 0, \alpha > 0, \nu > 0, \quad (2.12)$$

where  $\mathcal{K}_\nu(\cdot)$  is the modified Bessel function of second type, and  $\alpha$  and  $\nu$  refer to the *scaling and smoothness parameters* respectively. From the inversion formula (see p. 46 *ibid.*), one sees that

$$\mathcal{F}K(\xi) = (2\pi)^d \mathcal{F}^{-1}K(\xi) = (2\pi)^d f(|\xi|) \quad \text{with} \quad f(\xi) = \frac{2^{\nu-1} \phi \Gamma(\nu + d/2) \alpha^{2\nu}}{\pi^{d/2} (\alpha^2 + |\xi|^2)^{\nu+d/2}}, \quad \xi \in \mathbb{R}^d.$$

Comparing the above expression with (2.11), we see that the class of Bessel kernels  $f_s$ , with  $s > d - 2$  and  $d \geq 3$ , includes the Matérn class (2.12) as a special case under the following choice of parameters:

$$\alpha = 1, \quad \nu = (s - d)/2, \quad \text{and} \quad \phi = 2^{(2-d-s)/2} \pi^{-d/2} \Gamma(s/2)^{-1}.$$

Note that the requirement of the smoothness parameter  $\nu > 0$  for the Matérn class corresponds to the case of the Bessel kernel with  $s > d$ .

The following proposition shows what conditions (1.17), (1.13a), and (1.18) reduce to for the Bessel kernel as the correlation function in terms of its parameters.

**Proposition 2.12** (Bessel kernel as correlation function). *If the correlation function  $f$  is given by the Bessel kernel  $f_s(\cdot)$  with  $s > 0$  defined in Example 2.10, then*

$$\Upsilon_\alpha(0) = \frac{\Gamma\left(\frac{d}{2} - 1 + \alpha\right) \Gamma\left(\frac{s-d}{2} + 1 - \alpha\right)}{2^d \pi^{d/2} \Gamma(d/2) \Gamma(s/2)} \quad \text{for all } s > d - 2(1 - \alpha) > 0, \quad (2.13)$$

and in particular when  $\alpha = 0$ , (2.13) simplifies to the following:

$$\Upsilon(0) = \frac{\Gamma\left(\frac{2+s-d}{2}\right)}{2^{d-1} \pi^{d/2} (d-2) \Gamma(s/2)} \quad \text{for all } s > d - 2 > 0. \quad (2.14)$$

In addition,

$$\mathcal{H}_\alpha(t) < \infty \quad \forall t > 0 \quad \Longleftrightarrow \quad 0 < \alpha < \frac{1}{2} - \frac{(d-s)_+}{4} \quad \text{and} \quad s > d - 2 > 0, \quad (2.15)$$

where  $a_+ := \max(a, 0)$ . Moreover, for  $\alpha \in (0, 1/2)$ , we have the following asymptotic behavior of  $\mathcal{H}_\alpha(t)$  at  $t \rightarrow 0$ :

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2} \Gamma((d-s)/2)}{((s-d)/2 + 1 - 2\alpha) \Gamma(d/2)} t^{(s-d)/2 + 1 - 2\alpha} \\ \quad - \frac{\pi^{d/2} \Gamma((s-d)/2)}{(1-2\alpha) \Gamma(s/2)} t^{1-2\alpha} \\ \quad + O\left(t^{(s-d)/2 + 2(1-\alpha)}\right), & \begin{array}{l} d-2 < s < d \\ \text{and} \\ \alpha < \frac{1}{2} - \frac{1}{4}(d-s) \end{array} & (2.16\text{-a}) \\ \\ \frac{\pi^{d/2}}{(1-2\alpha) \Gamma(d/2)} t^{1-2\alpha} \log\left(\frac{1}{t}\right) \\ \quad + \frac{\pi^{d/2} (1 - (1-2\alpha) [\psi(d/2) + 2\gamma])}{(1-2\alpha)^2 \Gamma(d/2)} t^{1-2\alpha} \\ \quad + O(t^2 \log(t)), & s = d & (2.16\text{-b}) \\ \\ \frac{\pi^{d/2} \Gamma((s-d)/2)}{(1-2\alpha) \Gamma(s/2)} t^{1-2\alpha} + O(t^{(s-d)/2 + 1 - 2\alpha}), & d < s < d+2 & (2.16\text{-c}) \\ \\ \frac{\pi^{d/2}}{(1-2\alpha) \Gamma(d/2 + 1)} t^{1-2\alpha} + O(t^{2(1-\alpha)} \log(t)), & s = d+2 & (2.16\text{-d}) \\ \\ \frac{\pi^{d/2}}{(1-2\alpha) \Gamma(d/2 + 1)} t^{1-2\alpha} + O(t^{2(1-\alpha)}), & s > d+2 & (2.16\text{-e}) \end{cases} \quad (2.16)$$

where  $\psi(x) = \frac{d}{dx} \log \Gamma(x)$  refers to the digamma function and  $\gamma \approx 0.57721$  to Euler's constant; see, e.g., 5.2.2 and 5.2.3 on p. 136 of [21].

*Proof.* By the spherical coordinate integration formula and (2.11), for all  $\alpha \in [0, 1)$ ,

$$\Upsilon_\alpha(0) = (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{d\xi}{|\xi|^{2(1-\alpha)} (1 + |\xi|^2)^{s/2}} = (2\pi)^{-d} C_d \int_0^\infty \frac{r^{d-1}}{r^{2(1-\alpha)} (1 + r^2)^{s/2}} dr,$$

where  $C_d := \frac{2\pi^{d/2}}{\Gamma(d/2)}$ . Now by the change of variables  $z = r^2/(1+r^2)$ , we can evaluate the above integral explicitly by transforming it to the Beta integral:

$$\begin{aligned} \int_0^\infty \frac{r^{d-1}}{r^{2(1-\alpha)}(1+r^2)^{s/2}} dr &= \frac{1}{2} \int_0^1 z^{d/2+\alpha-2} (1-z)^{(s-d)/2-\alpha} dz \\ &= \frac{\Gamma(d/2-1+\alpha) \Gamma((s-d)/2+1-\alpha)}{2\Gamma(s/2)}, \end{aligned}$$

which is finite provided that  $s > d - 2(1-\alpha) > 0$ . This proves (2.13) and from this, we easily deduce (2.14) by letting  $\alpha = 0$  in (2.13) and by applying the formula  $\Gamma(z+1) = z\Gamma(z)$ , which holds for  $z \in \mathbb{C}$  such that  $\Re(z) > 0$ .

It remains to prove (2.16), which then implies (2.15). From (1.19) and by the spherical coordinate integration formula, for all  $t > 0$ ,

$$\begin{aligned} \mathcal{H}_\alpha(t) &= \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi \frac{\exp(-r|\xi|^2)}{(1+|\xi|^2)^{s/2}} \\ &= C_d \int_0^t dr r^{-2\alpha} \int_0^\infty dz \frac{\exp(-rz^2)}{(1+z^2)^{s/2}} z^{d-1} \\ &= \frac{C_d}{2} \int_0^t dr r^{-2\alpha} \int_0^\infty du \exp(-ru) (1+u)^{-s/2} u^{d/2-1} \\ &=: \frac{C_d \Gamma(d/2)}{2} \int_0^t dr r^{-2\alpha} I(r) = \pi^{d/2} \int_0^t dr r^{-2\alpha} I(r). \end{aligned}$$

By [21, 13.4.4 on p.326],  $I(r)$  is equal to the *confluent hypergeometric function*:

$$I(r) = U\left(\frac{d}{2}, \frac{2+d-s}{2}, r\right). \quad (2.17)$$

By 18.2.18 – 13.2.22 on p. 323 *ibid.*, we see that

$$I(r) = \begin{cases} \frac{\Gamma((d-s)/2)}{\Gamma(d/2)} r^{(s-d)/2} + \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r^{(s-d)/2+1}) & d-2 < s < d & 18.2.18, \\ -\frac{1}{\Gamma(d/2)} (\log(r) + \psi(d/2) + 2\gamma) + O(r \log(r)) & s = d & 18.2.19, \\ \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r^{(s-d)/2}) & d < s < d+2 & 18.2.20, \\ \frac{1}{\Gamma(d/2+1)} + O(r \log(r)) & s = d+2 & 18.2.21, \\ \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r) & s > d+2 & 18.2.22. \end{cases}$$

Then integrating the right-hand side of the above expressions against  $\pi^{d/2} r^{-2\alpha} dr$  over  $[0, t]$  gives the five cases in (2.16). This completes the proof of Proposition 2.12.  $\square$

Similarly, one can use the Bessel kernel as the spectral density. In this case, we have the following proposition:

**Proposition 2.13** (Bessel kernel as spectral density). *Suppose that the spectral density  $\hat{f}$  is given by the Bessel kernel  $f_s(\cdot)$  defined in Example 2.10, or equivalently (see (2.11)), suppose that  $f(x) = (1+|x|)^{-s/2}$  for  $s > 0$ . Then*

$$\Upsilon_\alpha(0) = \frac{\Gamma(1-\alpha) \Gamma(\alpha-1+s/2)}{2^{2d} \pi^{3d/2} \Gamma(d/2) \Gamma(s/2)} \quad \text{for all } s > 2(1-\alpha) > 0, \quad (2.18)$$

and in particular when  $\alpha = 0$ , (2.18) simplifies to the following:

$$\Upsilon(0) = \frac{2^{1-2d}\pi^{-3d/2}}{(s-2)\Gamma(d/2)} \quad \text{for all } s > 2. \quad (2.19)$$

In addition,

$$\mathcal{H}_\alpha(t) < \infty \quad \forall t > 0 \quad \Longleftrightarrow \quad 0 < \alpha < \frac{1}{2} \quad \text{and} \quad s > 0. \quad (2.20)$$

Moreover, for  $\alpha \in (0, 1/2)$ , we have the following asymptotic

$$\mathcal{H}_\alpha(t) \sim \frac{t^{1-2\alpha}}{1-2\alpha}, \quad \text{as } t \downarrow 0. \quad (2.21)$$

*Proof.* By similar arguments as Proposition 2.12, we have that

$$\begin{aligned} \Upsilon_\alpha(0) &= (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{f_s(\xi)}{|\xi|^{2(1-\alpha)}} d\xi = (2\pi)^{-2d} \int_{\mathbb{R}^d} \frac{\widehat{f_s}(\xi)}{|\xi|^{d-2(1-\alpha)}} d\xi \\ &= (2\pi)^{-2d} \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty \frac{r^{d-1}}{(1+r^2)^{s/2} r^{d-2(1-\alpha)}} dr = \frac{\Gamma(1-\alpha) \Gamma(\alpha-1+s/2)}{2^{2d}\pi^{3d/2}\Gamma(d/2)\Gamma(s/2)}, \end{aligned}$$

which is finite provided  $s > 2(1-\alpha)$ . This proves both (2.18) and (2.19). As for (2.20),

$$\begin{aligned} \mathcal{H}_\alpha(t) &= \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi f_s(\xi) \exp(-r|\xi|^2) \\ &= (2\pi)^{-d} \pi^{d/2} \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi \widehat{f_s}(\xi) \exp\left(-\frac{|\xi|^2}{4r}\right) r^{-d/2} \\ &= (2\pi)^{-d} \pi^{d/2} \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi (1+|\xi|^2)^{-s/2} \exp\left(-\frac{|\xi|^2}{4r}\right) r^{-d/2} \\ &= (2\pi)^{-d} \pi^{d/2} \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^t dr r^{-2\alpha-d/2} \int_0^\infty dz z^{d-1} (1+z^2)^{-s/2} \exp\left(-\frac{z^2}{4r}\right), \end{aligned}$$

where we have used Plancherel's theorem and the following identities:

$$\mathcal{F}(\exp(-|\cdot|^2))(\xi) = \pi^{d/2} \exp(-4^{-1}|\xi|^2) \quad \text{and} \quad \mathcal{F}(f(a\cdot))(\xi) = a^{-d} \mathcal{F}f(\xi/a).$$

Then, by the same arguments as Proposition 2.12,

$$\mathcal{H}_\alpha(t) = 2^{-d} \int_0^t dr r^{-2\alpha-d/2} I\left(\frac{1}{4r}\right), \quad \text{with } I(r) = U\left(\frac{d}{2}, \frac{2+d-s}{2}, r\right),$$

where  $U$  is given in (2.17). As  $r \rightarrow \infty$ ,  $I(r) \sim r^{-d/2}$  thus as  $r \rightarrow 0$ ,  $I(\frac{1}{4r}) \sim (4r)^{d/2}$  (see 13.2.6 on p. 322 of [21]). Hence, the above integral behave as follows:

$$\mathcal{H}_\alpha(t) \sim \int_0^t dr r^{-2\alpha-d/2} (4r)^{d/2} = \int_0^t dr r^{-2\alpha} = \frac{t^{1-2\alpha}}{1-2\alpha},$$

provided  $\alpha \in (0, 1/2)$ . This proves both (2.20) and (2.21).  $\square$

The necessity of the finiteness of  $\Upsilon(0)$  excludes the Riesz kernel as a choice for the spectral density. However, we can still construct a Riesz-type kernel which has polynomial growth at the origin and polynomial decay at infinity, but with different rates, using Propositions 2.12 and 2.13. This Riesz-type kernel gives another example of a kernel that is easily verifiable to be permissible under the conditions of our Theorem 1.4 above, while being very difficult to verify using (2.4); see Example 2.14 for more details.

**Example 2.14** (Riesz-type kernel). For  $s_1, s_2 \in (0, d)$ , let  $f_{s_1}$  and  $f_{s_2}$  be Bessel kernels as in Example 2.10. Define

$$r(x) := f_{s_1}(x) + \widehat{f_{s_2}}(x) \quad \text{or equivalently} \quad \widehat{r}(\xi) := \widehat{f_{s_1}}(x) + f_{s_2}(x).$$

It is easy to see that  $r(\cdot)$  is both non-negative and non-negative definite which follows immediately from the linearity of the Fourier transform and the fact that the Bessel kernel is both non-negative and non-negative definite. Also, we easily deduce from properties (2) - (4) in Example 2.10 that

$$r(x) \sim \begin{cases} |x|^{s_1-d} & |x| \rightarrow 0, \\ |x|^{-s_2} & |x| \rightarrow \infty, \end{cases} \quad \text{and} \quad \widehat{r}(\xi) \sim \begin{cases} |\xi|^{s_2-d} & |\xi| \rightarrow 0, \\ |\xi|^{-s_1} & |\xi| \rightarrow \infty. \end{cases}$$

Propositions 2.12 and 2.13 imply that

$$\Upsilon_\alpha(0) = \frac{\Gamma\left(\frac{d}{2} - 1 + \alpha\right) \Gamma\left(\frac{s_1-d}{2} + 1 - \alpha\right)}{2^d \pi^{d/2} \Gamma(d/2) \Gamma(s_1/2)} + \frac{\Gamma(1 - \alpha) \Gamma\left(\alpha - 1 + \frac{s_2}{2}\right)}{2^{2d} \pi^{3d/2} \Gamma(d/2) \Gamma(s_2/2)} < \infty,$$

provided that

$$0 < d - 2(1 - \alpha) < s_1 < d \quad \text{and} \quad 0 < 2(1 - \alpha) < s_2 < d.$$

In contrast, it is not clear how to compute  $\mathcal{F}(\sqrt{\widehat{r}})$ ; see condition (2.4).

## 2.5 Weight functions

The admissible condition 1.7 is needed in this paper due to the following result:

**Proposition 2.15** (Proposition 2.1 of [24]). *For any admissible weight  $\rho$ , the operators on  $L_\rho^2(\mathbb{R}^d)$  defined by  $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$  can be extended to a  $C_0$ -semigroup on  $L_\rho^2(\mathbb{R}^d)$ . Moreover, if  $\tilde{\rho}$  is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\tilde{\rho}(x)} dx < \infty,$$

*then for any  $t > 0$ , the operators defined above are compact from  $L_\rho^2(\mathbb{R}^d)$  to  $L_{\tilde{\rho}}^2(\mathbb{R}^d)$ .*

Here are several examples of admissible weights (see Example in Section 2 of [24]):

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \tag{2.22}$$

**Remark 2.16.** The smaller the weight function  $\rho(\cdot)$  (not necessarily admissible) is, the larger the space  $L_\rho^2(\mathbb{R}^d)$  is. For example, one may choose  $\rho$  to be either a nonnegative function with compact support or the heat kernel itself  $G(1, \cdot)$ . In both cases,  $\rho$  is smaller than those in (2.22) (up to a constant). However, one can easily check that the admissible condition (1.7) excludes these two cases. However, the examples in the following Proposition 2.17 seem to be less obvious.

**Proposition 2.17.**  $\rho_b(\cdot)$  is admissible if and only if  $b \in (0, 1]$  where

$$\rho_b(x) := \exp\left(-|x|^b\right), \quad x \in \mathbb{R}^d, \quad \text{with } b > 0.$$

*Proof.* From Definition (1.1), we see that  $\rho_b$  is admissible if and only if for all  $T > 0$ ,

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^d} \int_{\mathbb{R}^d} (2\pi t)^{-d/2} e^{-\frac{|x-y|^2}{2t} + (|x|^b - |y|^b)} dy < \infty.$$

Denote the above integral by  $I(t, x)$ . We will use  $C$  to denote a generic constant that does not depend on  $(t, x)$ , which value may change at each occurrence.

We first assume that  $b \in (0, 1]$ . In this case,

$$|x|^b = |x - y + y|^b \leq (|x - y| + |y|)^b \leq |x - y|^b + |y|^b.$$

Hence,

$$I(t, x) \leq C \int_{\mathbb{R}^d} t^{-d/2} e^{-\frac{|x-y|^2}{2t} + |x-y|^b} dy = C \int_{\mathbb{R}^d} t^{-d/2} e^{-\frac{|y|^2}{2t} + |y|^b} dy = C \int_0^\infty t^{-d/2} e^{-\frac{r^2}{2t} + r^b} r^{d-1} dr.$$

Then by applying the change of variables  $r' = t^{-1/2}r$ ,

$$I(t, x) \leq C \int_0^\infty e^{-\frac{r'^2}{2} + t^{b/2} r'^b} r'^{d-1} dr' \leq C \int_0^\infty e^{-\frac{r'^2}{2} + T^{b/2} r'^b} r'^{d-1} dr' < \infty.$$

Next we assume that  $b > 1$ . We need to show that  $\rho_b$  is not admissible. Without loss of generality, we assume that  $d \geq 2$ . The case when  $d = 1$  is easier and can be proved in a similar way as the proof below. It suffices to show that

$$\lim_{r \rightarrow \infty} I(1/2, x_r) = \infty, \quad \text{where } x_r := (r, 0, \dots, 0) \in \mathbb{R}^d.$$

Without loss of generality, we may assume below that  $r \gg 2$ . Denote  $y = (y_1, \dots, y_d) = (y_1, y_*)$  with  $y_* \in \mathbb{R}^{d-1}$ . Using the subadditivity (resp. convexity) of  $(x + y)^{b/2}$  when  $b \in (1, 2]$  (resp.  $b > 2$ ), we see that

$$|y|^b = (y_1^2 + y_2^2 + \dots + y_d^2)^{b/2} \leq c|y_1|^b + (y_2^2 + \dots + y_d^2)^{b/2} = c|y_1|^b + c|y_*|^b, \quad c := 1 \wedge 2^{b/2-1}.$$

Hence,

$$\begin{aligned} I(1/2, x_r) &= C \int_{\mathbb{R}^d} e^{-(\sum_{i=2}^d y_i^2) - |y_1 - r|^2 + (r^b - |y|^b)} dy \\ &\geq C \int_{\mathbb{R}^{d-1}} dy_* \int_{\mathbb{R}} dy_1 e^{-|y_*|^2 - |y_*|^b - |y_1 - r|^2 + (r^b - c|y_1|^b)} \\ &= C \int_{\mathbb{R}^{d-1}} dy_* e^{-|y_*|^2 - c|y_*|^b} \int_{\mathbb{R}} dy_1 e^{-|y_1 - r|^2 + r^b - c|y_1|^b} \\ &= C \int_{\mathbb{R}} e^{-y^2 + r^b - c|y - r|^b} dy. \end{aligned}$$

Therefore,

$$I(1/2, x_r) \geq CK(r), \quad \text{with } K(r) := \int_0^r e^{-y^2 + r^b - c(r-y)^b} dy.$$

It suffices to show that  $\lim_{r \rightarrow \infty} K(r) = \infty$ , which is clear when  $b > 2$  because

$$K(r) \geq \int_{r/2}^r e^{-y^2 + r^b - c(r-y)^b} dy \geq \int_{r/2}^r e^{-r^2 + r^b - c(r/2)^b} dy = \frac{r}{2} \exp\left(\left(1 - 2^{-1-b/2}\right)r^b - r^2\right),$$

which blows up as  $r \rightarrow \infty$ . Hence, we may assume that  $b \in (1, 2]$ . In this case,  $c = 1$  and

$$K'(r) = e^{r^b - r^2} + b \int_0^r e^{-y^2 + r^b - (r-y)^b} \left( r^{b-1} - (r-y)^{b-1} \right) dy.$$

By the intermediate value theorem, we see that

$$r^{b-1} - (r-y)^{b-1} = (b-1)y\xi^{b-2}, \quad \text{for some } \xi \in [r-y, r],$$

and since  $b-1 \in (0, 1]$ , this implies that  $r^{b-1} - (r-y)^{b-1} \geq (b-1)y r^{b-2}$ . Hence,

$$\begin{aligned} K'(r) &\geq e^{r^b - r^2} + b(b-1) \int_1^r e^{-y^2 + r^b - (r-y)^b} y r^{b-2} dy \\ &\geq b(b-1) r^{b-2} \int_1^r e^{-y^2 + r^b - (r-1)^b} dy \geq b(b-1) r^{b-2} e^{r^b - (r-1)^b} \int_1^2 e^{-y^2} dy. \end{aligned}$$

Another application of the intermediate value theorem shows that  $r^b - (r-1)^b = b\xi^{b-1}$  with  $\xi \in [r-1, r]$ . Hence,  $r^b - (r-1)^b \geq b(r-1)^{b-1}$  and then

$$K'(r) \geq C r^{b-2} \exp \left( b(r-1)^{b-1} \right).$$

Hence, for  $r \gg 2$ ,  $K'(r)$  is positive and unbounded as  $r \rightarrow \infty$ . Therefore, this implies that  $K(r)$  blows up as  $r \rightarrow \infty$ , which completes the proof of Proposition 2.17.  $\square$

### 3 Moment estimates – Proof of Theorem 1.2

We first state some known results and prove a moment bound in Corollary 3.3.

**Theorem 3.1** (Theorem 1.2 [8]). *Suppose that*

- (i) *the initial deterministic measure  $\mu$  satisfies (1.3);*
- (ii) *the spectral measure  $\hat{f}$  satisfies Dalang's condition (1.15),*

*Then (1.1) has a unique random field solution starting from  $\mu$ . Moreover, the solution is  $L^2(\Omega)$  continuous and is adapted to the filtration  $\{\mathcal{F}_t\}_{t \geq 0}$ .*

**Theorem 3.2** (Theorem 1.7 [8]). *Under the assumptions of Theorem 3.1, for any  $t > 0$ ,  $x \in \mathbb{R}^d$  and  $p \geq 2$ , the solution to (1.1),  $u(t, x)$ , given by (1.5) is in  $L^p(\Omega)$  and*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(G(t, \cdot) * |\mu|)(x)] H(t; \gamma_p)^{1/2}, \quad (3.1)$$

*where  $\bar{\varsigma} = L_0/L_b$ ,  $\gamma_p = 32pL_b^2$  (see (1.4) for  $L_0$  and  $L_b$ ) and the function  $H(t; \gamma_p)$  is nondecreasing in  $t$  (see [8] for the expression of the function  $H$ ).*

**Corollary 3.3.** *Under the same setting as Theorem 3.2, if the two conditions in (1.13) hold (see also (1.14)), then*

$$\|u(t, x)\|_p \leq C_p \left( 1 + (G(t, \cdot) * |\mu|)(x) \right), \quad \text{for all } p \text{ such that } 1/p \in (64L_b^2\Upsilon(0), 1), \quad (3.2)$$

*where  $C_p = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$ .*

*Proof.* Lemma 2.5 of [9] gives one sufficient condition, namely,  $2\gamma_p\Upsilon(0) < 1$ , under which the function  $H((t; \gamma_p))$  is bounded in  $t$ . Therefore, by taking into account the expression of  $\gamma_p$  in Theorem 3.2, we see that as a direct consequence of (3.1), whenever

$$32pL_b^2 < \frac{1}{2\Upsilon(0)}, \quad (3.3)$$

we have the  $p$ -th moment bounded as given in (3.2).  $\square$

Now we are ready to prove Theorem 1.2.

*Proof of Theorem 1.2.* Under condition (iii), we can apply Fubini's Theorem and the moment bound (3.2) below to see that for some constant  $C > 0$  independent of  $t$ , which may vary from line to line, that

$$\begin{aligned} \mathbb{E} \left( \|u(t, \cdot; \mu)\|_\rho^2 \right) &\leq C \mathbb{E} \left[ \int_{\mathbb{R}^d} \left( 1 + (G(t, \cdot) * |\mu|)(x) \right)^2 \rho(x) dx \right] \\ &= C \int_{\mathbb{R}^d} \mathbb{E} \left[ \left( (G(t, \cdot) * (1 + |\mu|))(x) \right)^2 \right] \rho(x) dx \\ &= C \mathcal{G}_\rho(t; \mu^*) < \infty, \end{aligned}$$

where we recall that  $\mu^* = 1 + |\mu|$ . This proves Theorem 1.2.  $\square$

**Remark 3.4** (Restarted SHE). Recall that the Markov property of the solution to (1.1) implies that for any  $t \geq t_0 > 0$ ,

$$u(t + t_0, x; \mu) \stackrel{\mathcal{L}}{=} u(t, x; u(t_0, \cdot; \mu)) =: v(t, x), \quad (3.4)$$

where  $\mathcal{L}$  refers to the equality in law and we have used  $v(t, x)$  to simplify the notation. It is known that  $v$  satisfies the following restarted SPDE:

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}_{t_0}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, x) = u(t_0, x; \mu), & x \in \mathbb{R}^d, \end{cases} \quad (3.5)$$

where  $\dot{W}_{t_0}(t, x) := \dot{W}(t + t_0, x)$  denotes the time shifted noise, i.e.,

$$\int_0^t \int_{\mathbb{R}^d} W_{t_0}(ds, dy) = \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} W(ds, dy). \quad (3.6)$$

Under the conditions in (1.13), Theorem 3.2 and (3.4) imply immediately that

$$\|v(t, x)\|_q = \|u(t + t_0, x; \mu)\|_q \leq C_q \left( 1 + (G(t + t_0, \cdot) * |\mu|)(x) \right) = C_q J_0(t + t_0, x; 1 + |\mu|), \quad (3.7)$$

for all  $q \geq 2$  and  $t > 0$ , where the constant  $C_q$  does not depend on  $t$ . Moreover, under the assumptions of Theorem 1.2, we have  $v(0, \cdot) \in L_\rho^2(\mathbb{R}^d)$  a.s. and

$$\mathbb{E} \left( \|v(t, x)\|_\rho^2 \right) = \mathbb{E} \left( \|u(t + t_0, x; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t + t_0; \mu^*) < \infty. \quad (3.8)$$

## 4 A factorization lemma

In this section, we establish a factorization lemma with corresponding moment estimates; see Lemmas 4.2 and 4.4 below. This factorization lemma appeared in [10]; check also Section 5.3.1 of [11]. For  $\alpha \in (0, 1)$ ,  $t > 0$  and  $x \in \mathbb{R}^d$ , define formally

$$\begin{aligned} (F_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t-s)^{\alpha-1} G(t-s, x-y) f(s, y) \, ds dy \quad \text{and} \\ (Y_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t-s)^{-\alpha} G(t-s, x-y) f(s, y) W(ds, dy). \end{aligned} \quad (4.1)$$

For  $F_\alpha$ , the first step of the proof of [24, Theorem 3.1] showed the following proposition:

**Proposition 4.1.** *Let  $\rho$  and  $\tilde{\rho}$  be given as in condition (i) of Theorem 1.4 (see (1.22)). For any  $q > 2$ ,  $t_0 > 0$  and  $\alpha \in (q^{-1}, 2^{-1})$ , the operator  $F_\alpha$ , as an operator from  $L^q((0, t_0); L_{\tilde{\rho}}^2(\mathbb{R}^d))$  to  $L_\rho^2(\mathbb{R}^d)$ , is compact.*

As for  $Y_\alpha$ , we have the following two lemmas, which hold for both the non-restarted SHE ( $t_0 = 0$ ) and the restarted SHE ( $t_0 > 0$ ).

**Lemma 4.2.** *Suppose that  $\mu$  — the initial condition for  $u$  — satisfies (1.3) and that  $\hat{f}$  satisfies Dalang's condition (1.15). Suppose that for some  $\alpha \in (0, 1/2)$ ,  $\mathcal{H}_\alpha(t)$  defined in (1.19) is finite for all  $t > 0$ . Fix an arbitrary  $t_0 \geq 0$ . Let  $v(t, x)$  be the solution to the restarted SHE (3.5) and  $\tilde{W}_{t_0}$  be the time-shifted noise (see (3.6)) when  $t_0 > 0$  and let  $v = u$  when  $t_0 = 0$ . Then*

$$Y_v(s, y) := [Y_\alpha b(\circ, v(\cdot, \circ))](s, y) = \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0}(dr, dz) \quad (4.2)$$

has the following properties:

(1) for all  $q \geq 2$ ,  $s > 0$ ,  $y \in \mathbb{R}^d$ ,

$$\|Y_v(s, y)\|_q^2 \leq H(s + t_0; 32qL_b^2) J_0^2(s + t_0, y; \mu^*) \mathcal{H}_\alpha(s) < \infty, \quad (4.3)$$

where we remind the reader that  $\mu^* := 1 + |\mu|$  and we refer to Theorem 3.2 for the function  $H(t; \gamma)$ ;

(2) under both conditions in (1.13), if  $\mathcal{H}_\alpha(t)$  is finite for some  $\alpha \in (64L_b^2\Upsilon(0), 1/2)$ , then for any  $q$  with  $1/q \in (64L_b^2\Upsilon(0), \alpha)$ , the function  $H(t; 32qL_b^2)$  in (4.3) is uniformly bounded in  $t \geq 0$ , i.e.,  $\sup_{t \geq 0} H(t; 32qL_b^2) < \infty$ ;

(3) under both conditions in (1.13), if  $\mathcal{H}_\alpha(t)$  is finite for some  $\alpha \in (64L_b^2\Upsilon(0), 1/2)$ , then for any  $q$  with  $1/q \in (64L_b^2\Upsilon(0), \alpha)$  and for any nonnegative and  $L^1(\mathbb{R}^d)$ -function  $\rho$ , there exists a constant  $\Theta = \Theta(q, L_b, L_0, \alpha)$ , which does not depend on  $t$ , such that for  $t > 0$ ,

$$\mathbb{E} \left( \int_0^t \|Y_v(s, \cdot)\|_\rho^q ds \right) \leq \Theta \int_0^t [\mathcal{G}_\rho(s + t_0; \mu^*) \mathcal{H}_\alpha(s)]^{q/2} ds, \quad (4.4)$$

which is finite provided that

$$\int_0^t [\mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty. \quad (4.5)$$

**Remark 4.3.** Condition (4.5) is true for  $t_0 > 0$  because  $\mathcal{G}_\rho(t; \mu)$  is a continuous function for  $t > 0$  and  $\mathcal{H}_\alpha(s)$  is continuous and bounded for  $s \in [0, t]$  thanks to (1.19). However, when  $t_0 = 0$ , the situation is much more trickier. For example, when the initial condition is the delta initial condition, we have that

$$\mathcal{G}_{\tilde{\rho}}(t; \delta_0) = \int_{\mathbb{R}^d} G(t, x)^2 \rho(x) dx = G(2t, 0) \int_{\mathbb{R}^d} G(t/2, x) \rho(x) dx < \infty,$$

where one can obtain the second equality via (1.25). Hence, when  $s \rightarrow 0$ ,  $\mathcal{G}_{\tilde{\rho}}(s; \delta_0)$  blows up with a rate  $s^{-d/2}$ . On the other hand,  $\mathcal{H}_\alpha(s)$  goes to zero with a different rate. One needs to combine these two rates to check if condition (4.5) holds. By introducing  $t_0$  and restarting the heat equation, one can avoid this issue, that being the potential singularity of  $\mathcal{G}_{\tilde{\rho}}$  at  $s = 0$ .

*Proof.* In the proof, we use  $C$  to denote a generic constant that may change its value at each appearance.

(1) We first prove (4.3). By the Burkholder-Davis-Gundy inequality and Minkowski's integral inequality, we see that

$$\begin{aligned} \|Y_v(s, y)\|_q^2 &\leq C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(z_1, v(r, z_1))\|_q \\ &\quad \times f(z_1 - z_2) G(s-r, y-z_2) \|b(z_2, v(r, z_2))\|_q. \end{aligned}$$

Note that for the Lipschitz condition in (1.4), we have that

$$|b(x, v)| \leq |b(x, v) - b(x, 0)| + |b(x, 0)| \leq L_b |v| + L_0 \leq C(1 + |v|), \quad C := L_b \vee L_0.$$

We apply this and the moment bound (3.1) to  $\|b(z_i, v(r, z_i))\|_q$  above to see that

$$\begin{aligned} \|b(z_i, v(r, z_i))\|_q &\leq C \left(1 + \|v(r, z_i)\|_q\right) \\ &= C \left(1 + \|u(r + t_0, z_i)\|_q\right) \\ &\leq CH(r + t_0; 32qL_b^2) J_0(r + t_0, z_i; \mu^*) \\ &\leq CH(s + t_0; 32qL_b^2) J_0(r + t_0, z_i; \mu^*), \quad i = 1, 2, r \in (0, s), \end{aligned} \quad (4.6)$$

where in the last step, we have used the fact that  $H(t; \gamma)$  is a nondecreasing function; see Lemma 2.6 of [9]. Therefore, by denoting  $C_s := H(s + t_0; 32qL_b^2)$ ,

$$\begin{aligned} \|Y_v(s, y)\|_q^2 &\leq CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left( G(s-r, y-z_i) J_0(r + t_0, z_i; \mu^*) \right) \\ &= CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ &\quad \times f(z_1 - z_2) \prod_{i=1}^2 \left( G(s-r, y-z_i) G(r + t_0, z_i - \sigma_i) \right) \\ &= CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) G(s + t_0, y - \sigma_1) G(s + t_0, y - \sigma_2) \\ &\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(\frac{(r + t_0)(s-r)}{s + t_0}, z_i - \sigma_i \frac{r + t_0}{s + t_0} - \frac{s-r}{s + t_0} y\right) \end{aligned}$$

$$\begin{aligned} &\leq CC_s(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) G(s+t_0, y-\sigma_1) G(s+t_0, y-\sigma_2) \\ &\quad \times \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{(r+t_0)(s-r)}{s+t_0} |\xi|^2\right), \end{aligned}$$

where we have applied (1.25) and Plancherel's theorem. Hence,

$$\|Y_v(s, y)\|_q^2 \leq CC_s(2\pi)^{-2d} J_0^2(s+t_0, y; \mu^*) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{(r+t_0)(s-r)}{s+t_0} |\xi|^2\right).$$

Because the function

$$t_0 \mapsto \frac{r+t_0}{s+t_0} = 1 - \frac{s-r}{s+t_0} \quad \text{for } t_0 > 0,$$

is nondecreasing in  $t_0$  whenever  $s > r > 0$ , we can replace the two appearances of  $t_0$  in the exponent of the above inequality by zero to see that

$$\|Y_v(s, y)\|_q^2 \leq CC_s(2\pi)^{-2d} J_0^2(s+t_0, y; \mu^*) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right). \quad (4.7)$$

Furthermore, by symmetry of  $r(s-r)/s$  and the fact that  $r(s-r)/s \geq r/2$  for all  $r \in [0, s/2]$ , we see that the above double integral is bounded by

$$\begin{aligned} &\leq 2 \int_0^{s/2} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right) \\ &= 2^{2(1-\alpha)} \int_0^{s/4} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\ &\leq 2^{2(1-\alpha)} \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\ &= 2^{2(1-\alpha)} \mathcal{H}_\alpha(s). \end{aligned}$$

Plugging the above bound back to (4.7) proves (4.3).

(2–3) Part (2) is a direct consequence of Theorem 3.2. It remains to prove (4.4). An application of Minkowski's inequality shows that

$$\mathbb{E}\left(\|Y_v(s, \cdot)\|_\rho^q\right) = \left\| \int_{\mathbb{R}^d} Y_v(s, y)^2 \rho(y) dy \right\|_{q/2}^{q/2} \leq \left( \int_{\mathbb{R}^d} \|Y_v(s, y)\|_q^2 \rho(y) dy \right)^{q/2}. \quad (4.8)$$

By the definition of  $\mathcal{G}_\rho(t; \mu)$  in (1.20) and by (4.3), we see that

$$\int_{\mathbb{R}^d} \|Y_v(s, y)\|_q^2 \rho(y) dy \leq C \mathcal{G}_\rho(s+t_0; \mu^*) \mathcal{H}_\alpha(s).$$

Plugging the above expression to the far right-hand side of (4.8) proves (4.4). Finally, the finiteness of the upper bound in (4.4) is guaranteed by condition (4.5). This completes the proof of Lemma 4.2.  $\square$

**Lemma 4.4** (Factorization lemma). *Suppose that  $\mu$  — the initial condition for  $u$  — satisfies (1.3) and  $\widehat{f}$  satisfies Dalang's condition (1.15). Assume that condition (1.19) is satisfied for some  $\alpha \in (0, 1/2)$ . Fix an arbitrary  $t_0 \geq 0$ . Let  $v(t, x)$  be the solution to the restarted SHE*

(3.5) and  $\dot{W}_{t_0}$  be the time-shifted noise (see (3.6)) when  $t_0 > 0$  and let  $v = u$  when  $t_0 = 0$ . Then the following factorization holds

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y_v(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz),$$

for all  $t > 0$  and  $x \in \mathbb{R}^d$ . As a consequence,

$$v(t, x) = [G(t, \cdot) * u(t_0, \cdot; \mu)](x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y_v](t, x), \quad \text{for all } t > 0 \text{ and } x \in \mathbb{R}^d. \quad (4.9)$$

*Proof.* The lemma is straightforward provided that one can switch the orders of stochastic and ordinary integrals:

$$\begin{aligned} & \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y_v(s, \cdot)](x) ds \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \\ & \quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0}(dr, dz) \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz) \end{aligned} \quad (4.10)$$

$$\begin{aligned} &= \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(z, v(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \\ &= \frac{\pi}{\sin(\alpha\pi)} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz), \end{aligned} \quad (4.11)$$

where the last step is the *Beta integral* which requires that  $\alpha \in (0, 1)$ . It remains to justify the two applications of the stochastic Fubini's theorem (see Theorem 5.30 of Chapter one in [12], or also [25] or Theorem 4.33 of [11]) in (4.10) and (4.11) in the following two steps.

**Step 1.** In this step, we justify the change of orders in (4.10). Note that  $t, x$  and  $s$  are fixed. It suffices to prove the following condition:

$$\begin{aligned} I_1 &:= \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ & \quad \times f(z_1 - z_2) \left( \prod_{i=1}^2 G(s-r, y-z_i) \right) \mathbb{E} \left( \prod_{i=1}^2 b(z_i, v(r, z_i)) \right) \\ &= \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y_v(s, y)\|_2^2 \\ &< +\infty. \end{aligned}$$

But this follows immediately from (4.3). Indeed,

$$\begin{aligned} \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y_v(s, y)\|_2^2 &\leq C \int_{\mathbb{R}^d} dy G(t-s, x-y) J_0^2(s+t_0, y; \mu^*) \mathcal{H}_\alpha(s) \\ &= C \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t-s, x-y) \iint_{\mathbb{R}^{2d}} \mu^*(dz_1) \mu^*(dz_2) \\ & \quad \times G(s+t_0, y-z_1) G(s+t_0, y-z_2). \end{aligned}$$

Now we bound the three heat kernels using (1.25) as follows:

$$\begin{aligned}
& G(t-s, x-y) \prod_{i=1}^2 G(s+t_0, y-z_i) \\
&= \frac{G(2(t-s), x-y)^2}{G(4(t-s), 0)} \prod_{i=1}^2 G(s+t_0, y-z_i) \\
&\leq 2^d \frac{G(2(t-s), x-y)^2}{G(4(t-s), 0)} \prod_{i=1}^2 G(2s+2t_0, y-z_i) \\
&= 2^{d[4(t-s)]^{d/2}} \prod_{i=1}^2 \left[ G(2s+2t_0, y-z_i) G(2(t-s), x-y) \right] \\
&= 2^{2d(t-s)^{d/2}} \prod_{i=1}^2 \left[ G(2(t+t_0), x-z_i) G\left(\frac{2(t-s)(s+t_0)}{t+t_0}, y - \frac{s+t_0}{t+t_0}(x-z_i)\right) \right] \\
&\leq 2^{2d(t-s)^{d/2}} \prod_{i=1}^2 \left[ G(2(t+t_0), x-z_i) G\left(\frac{2(t-s)(s+t_0)}{t+t_0}, 0\right) \right] \\
&\leq C_{t,s,t_0} \prod_{i=1}^2 G(2(t+t_0), x-z_i).
\end{aligned}$$

Therefore,  $I_1 \leq C_{t,s,t_0} \mathcal{H}_\alpha(s) J_0^2(2(t+t_0), x; \mu^*) < \infty$ .

**Step 2.** Similarly, as for (4.11), we need to show that

$$\begin{aligned}
I_2 &:= \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \left( \prod_{i=1}^2 G(t-r, x-z_i) \right) \mathbb{E} \left( \prod_{i=1}^2 b(z_i, v(r, z_i)) \right) < \infty.
\end{aligned}$$

By the Cauchy Schwartz inequality, (4.6) and because  $\alpha \in (0, 1/2)$ ,

$$\begin{aligned}
I_2 &\leq C \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left( G(t-r, x-z_i) J_0(r+t_0, z_i; \mu^*) \right) \\
&= C' \int_0^t dr (t-r)^{-\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left( G(t-r, x-z_i) J_0(r+t_0, z_i; \mu^*) \right).
\end{aligned}$$

Now by the same arguments as those leading to (4.3) (with  $2\alpha$  there replaced by  $\alpha$ ), we see that

$$I_2 \leq C J_0^2(t, x; \mu^*) \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right),$$

which is finite by (1.19) where we replace  $\alpha$  with  $\alpha/2$  and repeat the same steps right after (4.7). This completes the proof of Lemma 4.4.  $\square$

Finally, we characterize conditions (1.17) and (1.19) in the following lemma:

**Lemma 4.5.** *For all  $\alpha \in (0, 1/2]$ , we have the following properties:*

(1)  $(2\pi)^{-d} \mathcal{H}_\alpha(t) \leq \Gamma(1 - 2\alpha) \Upsilon_{2\alpha}(0)$  for all  $t > 0$  and hence

$$\Upsilon_{2\alpha}(0) < \infty \implies \mathcal{H}_\alpha(t) < \infty \text{ for all } t > 0; \quad (4.12)$$

(2)  $\lim_{t \rightarrow \infty} (2\pi)^{-d} \mathcal{H}_\alpha(t) = \Gamma(1 - 2\alpha) \Upsilon_{2\alpha}(0)$ ;

(3) if  $\Upsilon(0) < \infty$ , then the reverse implication of (4.12) holds.

*Proof.* We only need to consider the case when  $\alpha > 0$ . It is clear that the function  $\mathcal{H}_\alpha(t)$  is nondecreasing. Hence, part (2) implies part (1). As for part (2), by Fubini's theorem,

$$\lim_{t \rightarrow \infty} \mathcal{H}_\alpha(t) = \int_0^\infty dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) e^{-r|\xi|^2} = \Gamma(1 - 2\alpha) \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^{2(1-2\alpha)}} = C \Upsilon_{2\alpha}(0), \quad (4.13)$$

with  $C := \Gamma(1 - 2\alpha)(2\pi)^d$ . Now for part (3), for any  $t > 0$ , by splitting the  $dr$  integral in (4.13) into two parts, we see that

$$\begin{aligned} C \Upsilon_{2\alpha}(0) &= \int_0^\infty dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) = \mathcal{H}_\alpha(t) + I_\alpha(t), \quad \text{with} \\ I_\alpha(t) &= \int_t^\infty dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2). \end{aligned}$$

Notice that

$$I_\alpha(t) \leq t^{-2\alpha} \int_t^\infty dr \int_{\mathbb{R}^d} \widehat{f}(d\xi) e^{-r|\xi|^2} = t^{-2\alpha} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^2} e^{-t|\xi|^2} \leq t^{-2\alpha} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^2} = \frac{(2\pi)^d}{t^{2\alpha}} \Upsilon(0).$$

Therefore,

$$\Upsilon_{2\alpha}(0) \leq \frac{\mathcal{H}_\alpha(t)}{(2\pi)^d \Gamma(1 - 2\alpha)} + \frac{\Upsilon(0)}{\Gamma(1 - 2\alpha) t^{2\alpha}} < \infty, \quad \text{for all } t > 0,$$

which proves part (3). □

## 5 Tightness and construction – Proof of Theorem 1.4

### 5.1 Proof of part (a) of Theorem 1.4

We are now ready to prove part (a) of Theorem 1.4.

*Proof of Theorem 1.4 (a).* In this proof,  $u(t, x)$  refers to  $u(t, x; \mu)$ . Fix  $\tau > 0$  and let  $t_0 = \tau/2$ . Throughout the proof, we have  $t \geq \tau$ . Let  $v$  be the solution to (3.5) that is restarted from  $t - t_0$ . Then (see Figure 3 for an illustration)

$$v_t(s, x) \stackrel{\mathcal{L}}{=} u(s, x; u(t - t_0, \cdot; \mu)) \quad \text{for } s \geq 0 \text{ and } t \geq \tau. \quad (5.1)$$

According to Assumption (i), we can choose and fix some admissible weight function  $\tilde{\rho}$  such that (1.22) is satisfied. Hence, by Proposition 2.15, the following set

$$\mathcal{K}_1(\Lambda) := \left\{ (G(t_0, \cdot) * y(\cdot))(x) : \|y\|_{\tilde{\rho}} \leq \Lambda \right\} \quad \text{with } \Lambda > 0$$

is relatively compact in  $L^2_{\tilde{\rho}}(\mathbb{R}^d)$ .

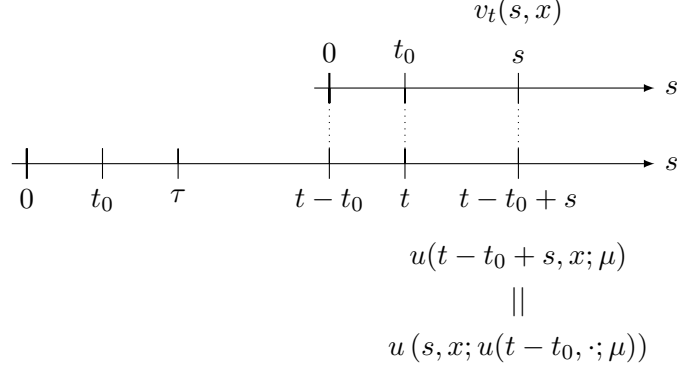


Figure 3: An illustration for the restarted SHE in (5.1).

Assumption (iii), i.e., (1.13), implies that the interval  $(64L_b^2\Upsilon(0), 1/2)$  is not empty. Moreover, Assumption (iv), i.e., (1.17), guarantees that there exists a constant  $\alpha$  in this interval, namely,  $64L_b^2\Upsilon(0) < \alpha < 1/2$ , such that (1.17) holds with  $\alpha$  replaced by  $2\alpha$ , i.e.,  $\Upsilon_{2\alpha}(0) < \infty$ . Now we can apply part (3) of Lemma 4.5, thanks to (1.13a), to see that  $\Upsilon_{2\alpha}(0) < \infty$  if and only if (1.19) holds. Therefore, both Lemmas 4.2 and 4.4 (more precisely part (3) of Lemma 4.2) are applicable. In particular, Lemma 4.4 ensures that the following factorization is well defined:

$$v(t_0, x) = \left( G(t_0, \cdot) * u(t - t_0, \cdot) \right)(x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y_v](t_0, x). \quad (5.2)$$

Part (3) of Lemma 4.2 shows that for any  $q$  in the following range,

$$64L_b^2\Upsilon(0) < \frac{1}{q} < \alpha < \frac{1}{2} \quad \left( \text{or equivalently } 2 < \frac{1}{\alpha} < q < \frac{1}{64L_b^2\Upsilon(0)} \right), \quad (5.3)$$

we can apply Proposition 4.1 to see that the set

$$\mathcal{K}_2(\Lambda) := \left\{ (F_\alpha h)(t_0, x) : \|h\|_{L^q((0, t_0); L_{\tilde{\rho}}^2(\mathbb{R}^d))} \leq \Lambda \right\}, \quad \text{with } \Lambda > 0,$$

is relatively compact in  $L_{\tilde{\rho}}^2(\mathbb{R}^d)$ . Now for any  $\Lambda > 0$ , define the set  $\mathcal{K}(\Lambda)$  as

$$\begin{aligned} \mathcal{K}(\Lambda) &:= \mathcal{K}_1(\Lambda) + \mathcal{K}_2(\Lambda) \\ &= \left\{ (G(t_0, \cdot) * y(\cdot))(x) + (F_\alpha h)(t_0, x) : \|y\|_{\tilde{\rho}} \leq \Lambda \quad \text{and} \quad \|h\|_{L^q((0, t_0); L_{\tilde{\rho}}^2(\mathbb{R}^d))} \leq \Lambda \right\}. \end{aligned}$$

Notice that from the factorization formula (5.2),

$$\begin{aligned} \mathbb{P}[v(t_0, \cdot) \notin \mathcal{K}(\Lambda)] &\leq \mathbb{P} \left[ \left( \int_0^{t_0} \|Y_v(s, \cdot)\|_{\tilde{\rho}}^q ds \right)^{1/q} > \frac{\pi\Lambda}{\sin(\alpha\pi)} \right] + \mathbb{P} \left[ \|u(t - t_0, \cdot)\|_{\tilde{\rho}} > \Lambda \right] \\ &=: I_1 + I_2. \end{aligned}$$

By Chebyshev's inequality and (1.21), we see that

$$I_2 \leq \frac{1}{\Lambda^2} \mathbb{E} \left( \|u(t - t_0, \cdot)\|_{\tilde{\rho}}^2 \right) \leq \frac{1}{\Lambda^2} \mathcal{G}_{\tilde{\rho}}(t - t_0; \mu^*).$$

Because  $\mathcal{G}_{\bar{\rho}}(t; \mu^*)$  is a continuous function for  $t > 0$ , and because it is also bounded at infinity, thanks to Assumption (ii) (see (1.23)), we have that

$$\mathcal{G}_{\bar{\rho}}(t - t_0; \mu^*) \leq \sup_{t \geq \tau} \mathcal{G}_{\bar{\rho}}(t - t_0; \mu^*) = \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) < \infty. \quad (5.4)$$

Therefore, we can bound  $I_2$  from above with a constant that does not depend on  $t \geq \tau$ , namely,

$$I_2 \leq \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) < \infty.$$

As for  $I_1$ , with the choice of  $\alpha$  and  $q$  in (5.3), one can apply Chebyshev's inequality and part (3) of Lemma 4.2 to see that

$$I_1 \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \mathbb{E} \int_0^{t_0} \|Y_v(s, \cdot)\|_{\bar{\rho}}^q ds \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \Theta \int_0^{t_0} (\mathcal{G}_{\bar{\rho}}(s + t - t_0; \mu^*) \mathcal{H}_{\alpha}(s))^{q/2} ds,$$

where the constant  $\Theta$  does not depend on  $t$ . As we have seen from above, since  $\Upsilon_{2\alpha}(0) < \infty$ , we can apply Lemma 4.5 to bound  $\mathcal{H}_{\alpha}(s)$  from above by the finite bound:  $(2\pi)^d \Gamma(1 - 2\alpha) \Upsilon_{2\alpha}(0)$ . Hence, together with (5.4), we obtain the following upper bound for  $I_1$  that is uniform in  $t \geq \tau$ :

$$I_1 \leq \frac{\sin^q(\alpha\pi) \Theta (2\pi)^{dq/2} t_0}{\Gamma(1 - 2\alpha)^{q/2} \pi^q \Lambda^q} \left( \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) \right)^{q/2} \Upsilon_{2\alpha}^{q/2}(0).$$

Combining these two upper bounds, we see that

$$\mathbb{P}[v(t_0, \cdot) \notin \mathcal{K}(\Lambda)] \leq \frac{\sin^q(\alpha\pi) \Theta (2\pi)^{dq/2} t_0}{\Gamma(1 - 2\alpha)^{q/2} \pi^q \Lambda^q} \left( \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) \right)^{q/2} \Upsilon_{2\alpha}^{q/2}(0) + \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) < \infty,$$

with the upper bound holding uniformly for all  $t \geq \tau$ . Hence, for any  $\epsilon > 0$ , by choosing  $\Lambda > 0$  big enough such that

$$\frac{\sin^q(\alpha\pi) \Theta (2\pi)^{dq/2} t_0}{\Gamma(1 - 2\alpha)^{q/2} \pi^q \Lambda^q} \left( \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) \right)^{q/2} \Upsilon_{2\alpha}^{q/2}(0) + \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu^*) < \epsilon,$$

we can ensure that

$$\mathbb{P}(u(t, \cdot) \in \mathcal{K}(\Lambda)) = \mathbb{P}(v(t_0, \cdot) \in \mathcal{K}(\Lambda)) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau,$$

which proves part (a) of Theorem 1.4.  $\square$

## 5.2 Proof of part (b) of Theorem 1.4

*Proof.* Fix an arbitrary  $\tau > 0$  and denote

$$U(T) := \frac{1}{T} \int_{\tau}^{T+\tau} \mathcal{L}(u(t, \cdot; \mu)) dt, \quad T > 0.$$

We claim that the family of laws  $U(T, \cdot)$  for  $T > 0$  is tight in  $L_{\rho}^2(\mathbb{R}^d)$ . Indeed, for any  $\epsilon \in (0, 1)$ , by part (a), there exists a compact set  $\mathcal{K} \in L_{\rho}^2(\mathbb{R}^d)$  such that (1.24) holds. This implies that

$$U(T)(\mathcal{K}) = \frac{1}{T} \int_{\tau}^{T+\tau} \mathcal{L}(u(t, \cdot; \mu))(\mathcal{K}) dt \geq \frac{1}{T} \int_{\tau}^{T+\tau} (1 - \epsilon) dt = 1 - \epsilon, \quad \text{for all } T > 0.$$

Let  $\{T_n\}_{n \in \mathbb{N}}$  be any deterministic sequence such that  $T_n \uparrow \infty$ . Since  $\{U(T_n)\}_{n \geq 1}$  is a tight sequence of measures, then there exists a subsequence  $\{U(T_{n_m})\}_{m \geq 1}$  that converges weakly to a measure,  $\eta$ , on  $L_{\rho}^2(\mathbb{R}^d)$  (e.g. see [3, Theorem 5.1]). Then one can apply the Krylov-Bogoliubov existence theorem (see, e.g., [11, Theorem 11.7]) to conclude that the measure  $\eta$  is an invariant measure for  $\mathcal{L}(u(t, \cdot; \mu))$ ,  $t \geq \tau$ . Finally, since  $\tau$  can be arbitrarily close to zero, one can conclude part (b) of Theorem 1.4.  $\square$

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